Epsilon 140/140Z/140Z G2/ 140LC User Manual



1 Description
2 Specification
3 Laser Safety
4 External Interface
5 Epsilon Control Application
6 Epsilon Operation
7 Maintenance

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Revision Index

Revision Number	Revision Date
1.4	11.Sep.2018
1.5	11.Mar.2020
1.6	19.Mar.2020
1.7	19.Mar.2020
1.8	10.Nov.2020
1.9	21.Dec.2020
2.0	11.Jun.2021

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Change Log

Page	Comment
1-5	Editorial changes
2-1	Clarification for analog video output for the 140Z G2
4-3	Removed 140Z G2
5-1	Added illustrations
5-2	Editorial changes
5-3	Editorial changes
5-4	Editorial changes
5-8	Box Size and additional basic feature description added.
5-9	Illustration updated
5-11	Editorial changes
5-11	Added active sensor illustration
5-12	Added False Color available palletes
5-13	Editorial changes
5-14	Added clarification for "Apply" and "Save and reset" buttons
5-15	Editorial changes
5-18	LRF illustration updated
5-23	Added video overlay description
5-24	Added KLV window description
6-1	Selecting Target Box Location And Size added
7-4	More detailed ilustration added
7-5	Editorial changes
7-6	Editorial changes

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Abbreviations

Abbreviation	Definition
FFC	Flat Field Correction
FOV	Field Of View
IR	Infrared
KLV	Key Length Value
LRF	Laser Range Finder
LWIR	Long-Wave Infrared
UAV	Unmanned Aerial Vehicle

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1.1 General

Epsilon 140/140Z/140LC/140Z G2 is a four sensor gyro-stabilized gimbal with direct drive torque motor technology providing stable video even at 30x optical zoom. The 140Z version has LWIR sensor with long-range continuous zoom providing up to 7.7° vertical FOV. All Epsilon 140/140Z/140LC/140Z G2 has state of the art onboard image processing capability. The onboard processing allows target tracking and eliminates any communication latency issues. It's processed video reduces the data rate and requires lower bandwidth datalinks, which is of an extreme importance for modern UAVs. The video stream along with snapshots can be stored in HD quality onboard the Epsilon 140/140Z/140LC/140Z G2.

Epsilon 140/140Z/140LC/140Z G2 is capable of hands-free tracking of moving objects. The operator can allocate the target and zoom in, while the onboard processor will hold the target in the center of the video screen, regardless of the aircraft movement direction. This is a fundamental feature for long-range surveillance, monitoring applications and law enforcement applications. Target tracking is paramount in applications where the aircraft is moving, and when the object of interest is moving relative to the ground.

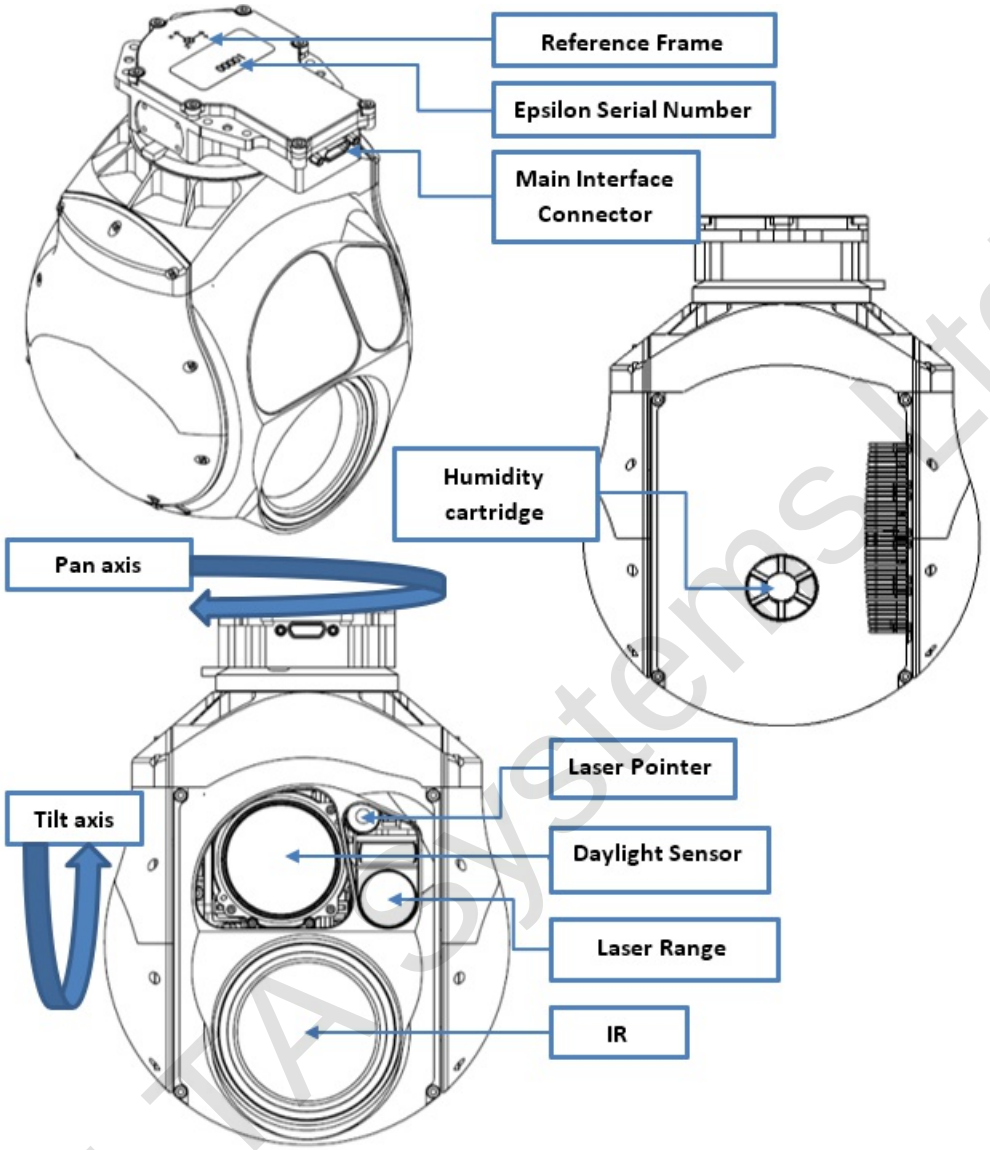
You can operate the Epsilon 140/140Z/140LC/140Z G2 in any conditions as it's IP64 rated. It's environmentally sealed housing will protect electronics from dust and heavy rain. The dome design is very robust and optimized for demanding applications were the gimbal needs to handle shock and rapid accelerations. Epsilon 140/140Z/140LC/140Z G2 will handle routine parachute recoveries, net recoveries, belly landings and catapult launch with ease.

Electronic stabilization with roll correction, removes the high frequency jitter often found on small unmanned aircraft platforms. This significantly reduces the operator workload and allows concentrating on the mission objective.

Epsilon will automatically hold the video centered on the scene, independent of the aircraft movement, even at tele-zoom setting.

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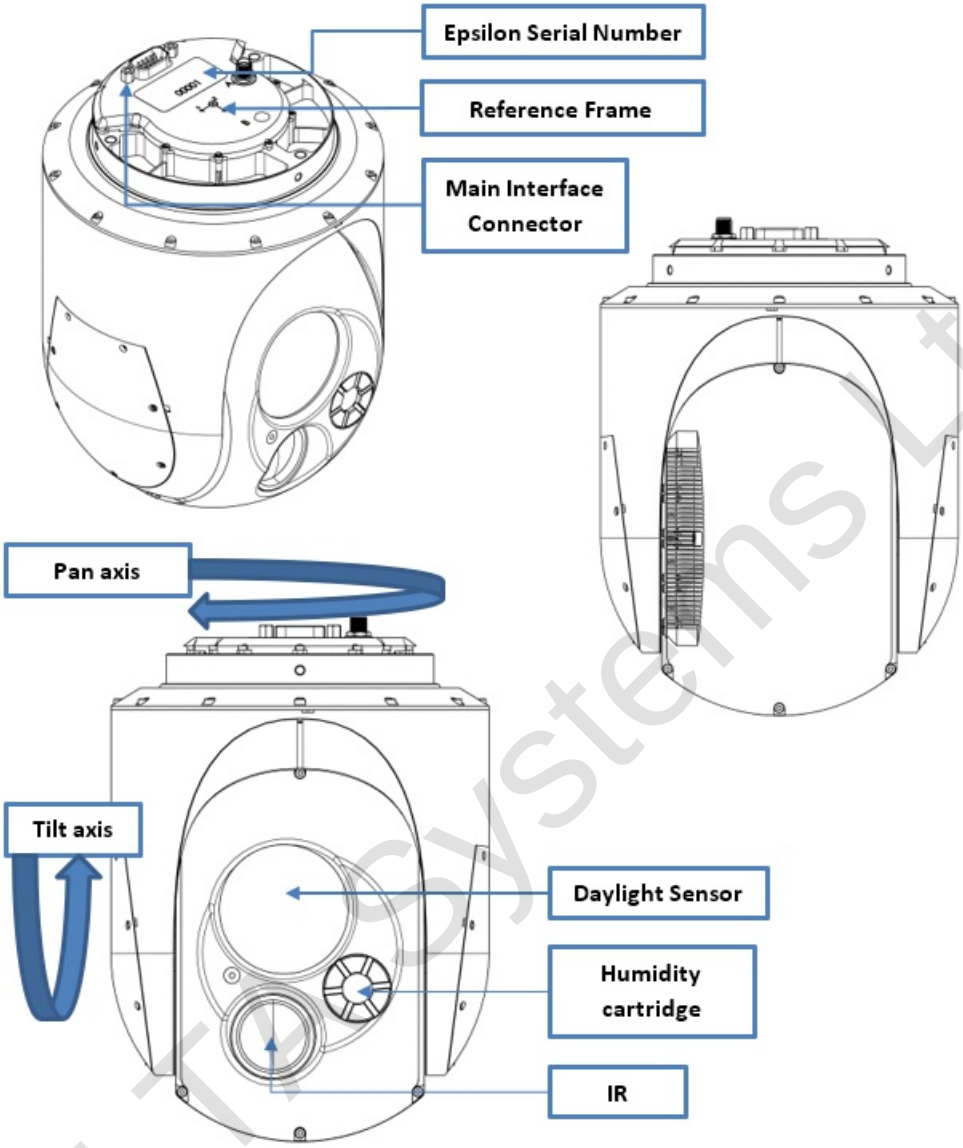
Epsilon 140Z overview



Epsilon 140Z overview

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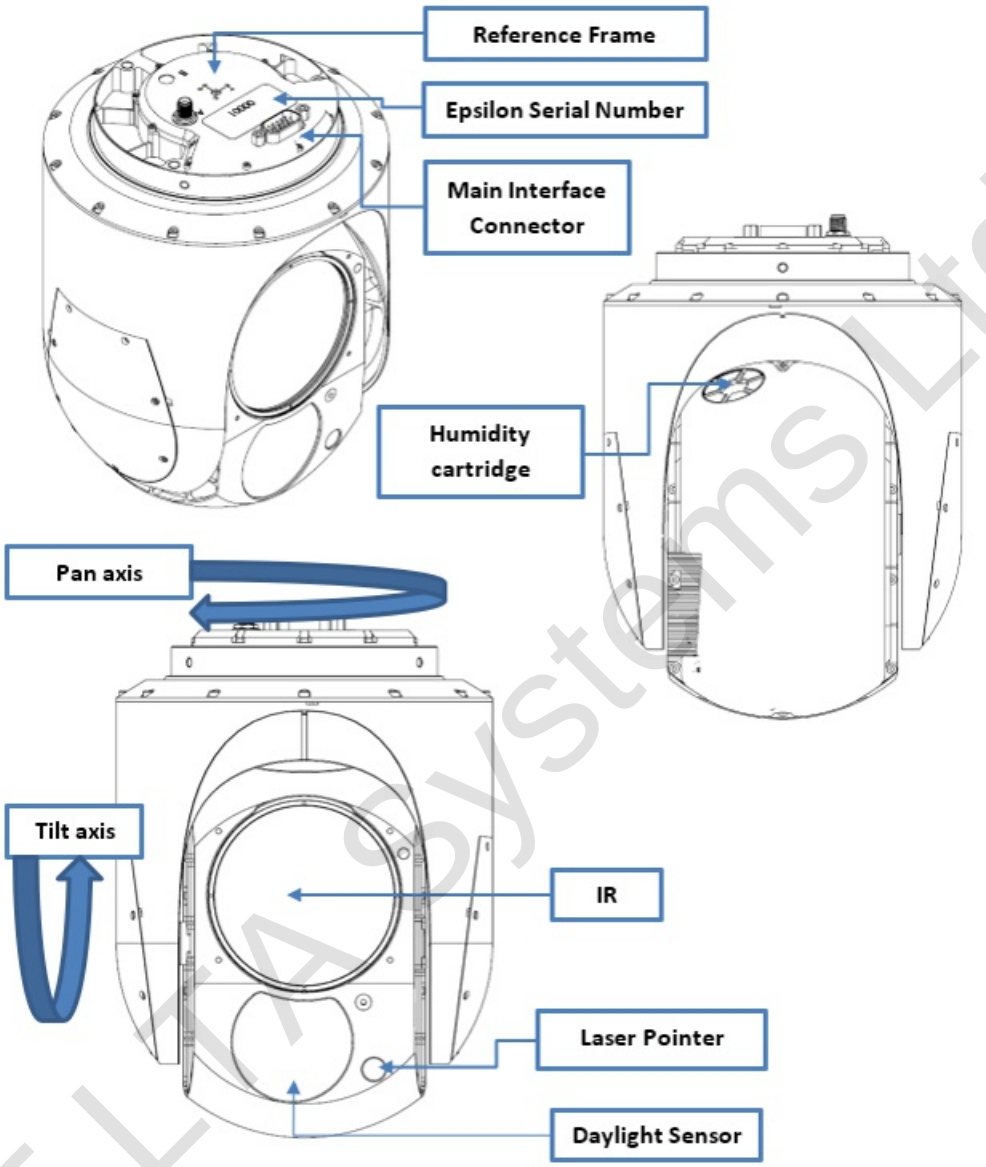
140LC overview



140LC overview

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140Z G2 overview



140Z G2 overview

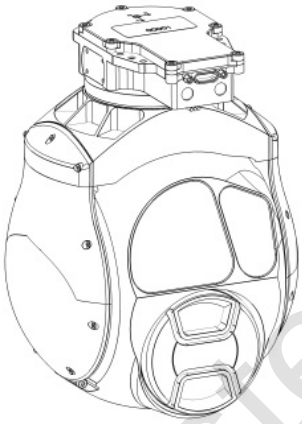
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1.2 Epsilon Camera Cap

Lenses are one of the most important parts of the camera to the quality of the image. The lens cap is designed to protect the lens. Lens cap provides protection from scratches and minor collisions for the camera. Camera cap/cover should be used at all times, when Epsilon 140/140Z/140Z G2 is not used for operations. Epsilon 140LC doesn't utilize a cap for lenses, but still special care should be taken when storing Epsilon 140LC to ensure that lenses are protected.

1.2.1 Epsilon 140 IR Camera Lens Cap

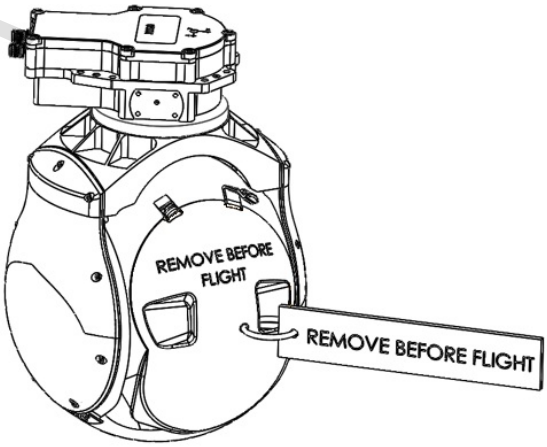
Epsilon 140 camera lens cap protects only IR camera.



Epsilon 140 Camera Lens Cap installed

1.2.2 Epsilon 140Z Camera Cover

Epsilon 140Z camera cap protects the entire front of Epsilon 140Z Gimbal. Camera cover holds in place with 2 magnets and 2 fasteners.

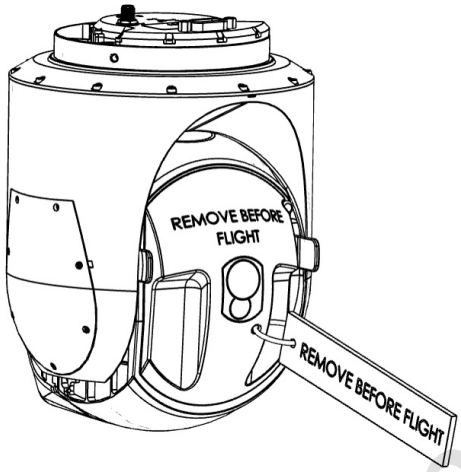


Epsilon 140Z Camera Cover installed

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1.2.3 Epsilon 140Z G2 Camera Cover

Cover is held in place with magnets and is easily removed by holding it and pulling it away from the camera itself.



Epsilon 140Z G2 Camera Cover installed

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2.1 System Specification

System Specifications	
Stabilization	<150 µrad
Size	**140/140Z:** 140 mm diameter x 189 mm **140LC:** 140 mm diameter x 195 mm **140Z G2:** 140 mm diameter x 212 mm
Weight	**140:** 1570 grams **140Z:** 1770 grams **140LC:** 1350 grams **140Z G2:** 1408 grams
Environmental Protection	IP64 rated
Operating Temperature	-40 to +55° C
Rotation Limits	**140/140Z:** 360° continuous pan, -80° to +35° elevation **140LC/140Z G2:** 360° continuous pan, -35° to +67° elevation
Slew rate	120 deg/sec
Power	**140/140Z:** 20 W typical/ 50W peak **140LC/140Z G2:** 17 W typical/ 75W peak
Input Voltage	24 Volts
Digital Video Out	Yes
Analog Video Out	**140/140Z/140LC:** Analog PAL or NTSC **140Z G2:** No
Control Interface	RS 232, Ethernet
Epsilon Control Application	Yes (Windows 7, 10)

HD Daylight sensors 140/140Z/140LC		IR Night sensor	
Sensor name	Hitachi DISC120R	**Sensor name**	FLIR TAU2
Type	EO sensor	**Type**	Uncooled VOx MicrobolometerIR
Global shutter	Yes	**Lens**	**140:** 60 mm fixed lens **140Z:** 3.3x zoom lens 18 – 60 mm **140LC:** 25 mm fixed lens **140Z G2:**5x zoom lens 15-75 mm
Vertical FOV	37.9°-1.3°	**Vertical FOV**	**140:** 7.7° **140Z:** 25.4° – 7.7° **140LC:** 18.9° **140Z G2:** 30.9° – 6.7°
Resolution	HD 1280x720 (output resolution 960x720)	**Resolution**	640x512
Optical zoom	30x	**Frame rate**	30 fps / 9 fps
Digital zoom	4x	**Digital zoom**	4x

HD Daylight sensors 140Z G2	
Sensor name	Sony FCB-EV7520
Type	EO sensor
Vertical FOV	24.8°-0.8°
Resolution	1920 x 1080
Optical zoom	30x
Digital zoom	4x

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Onboard Image Processor Specification	
Object tracking	Yes
Software stabilization with rotation correction	Yes
HD Video output	720p HD output / 1080p full HD for Epsilon 140Z G2
H.264 encoding	Yes
Onboard Video Recording / Snapshots	32 Gb onboard memory
Moving Target Indicator	Yes
Video Enhancement	Yes

Laser(if installed)			
Laser Rangefinder (140/140Z)		**Laser Pointer[1] (140/140Z/140Z G2)**	
Type	Diode laser 1.5 µm	Type	830 nm
Class	I (Eyesafe)	Class	III b
Range[2]	Up to 5000 m	Power	30 mW
Accuracy	<1m		

[1] Please read Laser Safety section. DO NOT point laser beam into the direction of someone's eyes.

[2] Maximum range depends on weather conditions, target surface and flying speed.

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3.1 Important

PLEASE READ CAREFULLY!

Caution must be taken when using a laser device due to the potential harmful effects of staring into a laser beam. **NEVER** stare into a laser beam directly or indirectly. **DO NOT** point laser beam into the direction of someone's eyes. Avoid direct eye exposure to the beam.

3.2 Class I

A Class 1 laser is safe under all conditions of normal use. The maximum permissible exposure cannot be exceeded when viewing a laser with the naked eye or with the aid of typical magnifying optics (e.g. telescope or microscope). It is important to realize that certain lasers classified as Class 1 may still pose a hazard when viewed with a telescope or microscope of sufficiently large aperture. An unsafe power level may be collected by a magnifying optic with larger aperture.

3.3 Class IIIb

A Class IIIb laser may cause damage if the beam enters the eye directly. Lasers in this class can cause permanent eye damage. A diffuse reflection is generally not hazardous but specular or mirror like reflections can be just as dangerous as direct exposures. Protective eyewear is recommended when direct beam viewing of Class IIIb lasers may occur.



More Information can be found here: **http://www.lasersafetyfacts.com/**

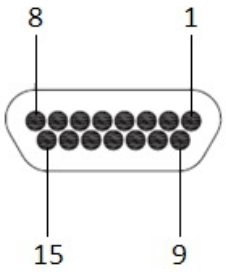
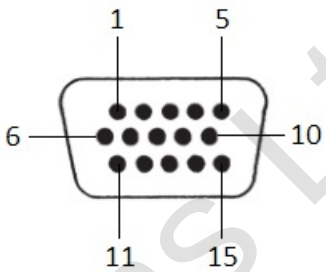
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4.1 Common Description

Epsilon external interface connectors

Epsilon Pinout	
140/140Z	**140LC/140Z G2**
Epsilon 140/140Z gimbals utilize AXON micro D female MDA2L15SFSB interface connector, which mates with AXON connector MDA2L15PFSB.	Epsilon 140LC and 140Z G2 gimbals utilize D-sub 9 high density male interface connector, which mates with D-sub 9 high density female connector.
	

Epsilon 140/140Z/140LC/140Z G2 uses 24V DC power with peak current up to 2A for Epsilon 140 and 140Z and 3A for Epsilon 140LC/140Z G2. The appropriate DC power supply must be used to provide 24V and 2A/3A power. Epsilon supports RS232 serial communication as well as Ethernet for main control. Both interfaces are developed to use direct PC connection and remote radio connection via RF data links. Serial port baud rate is 57600. As for video, there are two options for video output:

1. H.264 digital video via Ethernet (IP video), discussed further in **4.2 - Digital Video Output**
2. Composite analog video (Pal / NTSC), discussed further in **4.3 - Analog Video Output**

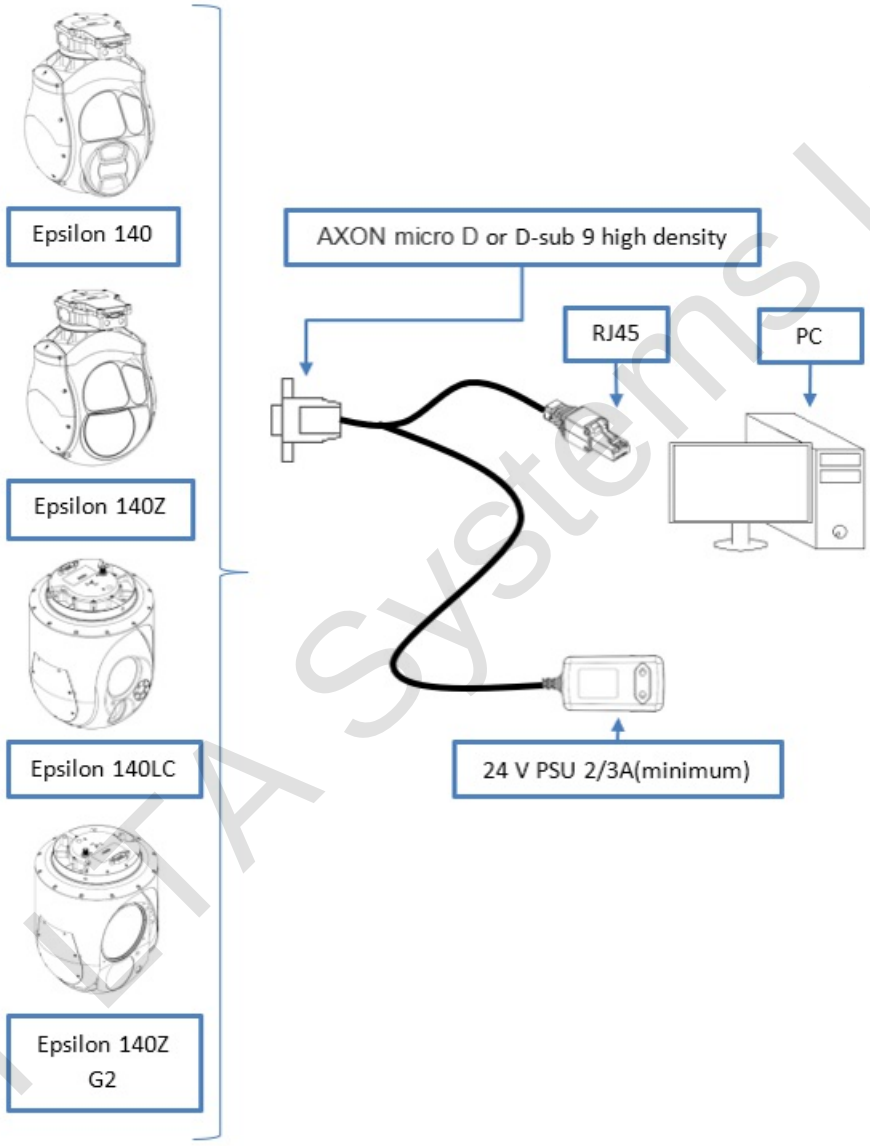
Main connector pinout is given the table below. Please note that it is for both styles of connector, the AXON micro D and D-sub 9 high density male.

Pinout description		
Pin	**Name**	**Description**
1	GND	Ground
2	+24V	+24V, 2A max current (140/140Z), 3A (140LC/ 140Z G2)
3	Tx-	Ethernet Tx- (RJ-45 pin 2)
4	Rx+	Ethernet Rx+ (RJ-45 pin 3)
5	AV	Composite Video signal / not connected on the 140Z G2
6	Tx	Connection to RS232 (Gimbal Tx)
7	nc	not connected
8	nc	not connected
9	GND	Ground
10	+24V	+24V, 2A max current (140/140Z), 3A (140LC/ 140Z G2)
11	Tx+	Ethernet Tx+ (RJ-45 pin 1)
12	Rx-	Ethernet Rx- (RJ-45 pin 6)
13	AVGND	Ground for analog video signals / not connected on the 140Z G2
14	Rx	Connection to RS232 (Gimbal Rx)
15	nc	not connected

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4.2 Digital Video Output

Epsilon 140 series camera has an integrated Onboard Image processing unit with digital video output feature. The H.264 codec is commonly used for HD video encoding. The video stream is transferred via UDP using Ethernet interface. The same Ethernet interface is also used to control the unit. The RS232 serial can still be used instead of Ethernet to control the unit. The default Epsilon IP address is 192.168.10.100. To change IP address, refer to section **5.7 - Common Settings**. For detailed digital video settings, refer to section **5.8 - Video Settings**.

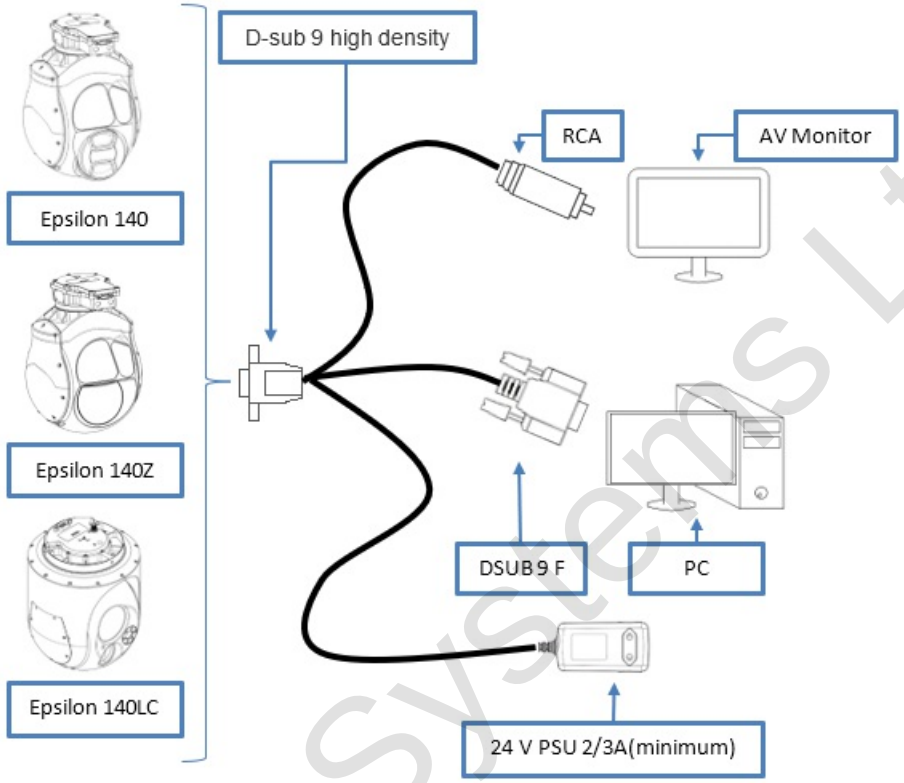


Typical Epsilon connection for digital video

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4.3 Analog Video Output

The second option for video output is composite analog video. Analog video output is available for Epsilon 140/ 140Z/ 140LC. In this case video can be viewed by analog video input devices. Analog video can be configured for PAL or NTCS. For detailed video settings, refer to **section 5.8.** Typical connection is shown below.



Typical Epsilon connection for analog video

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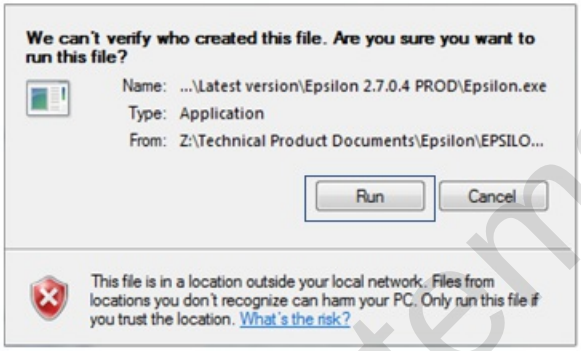
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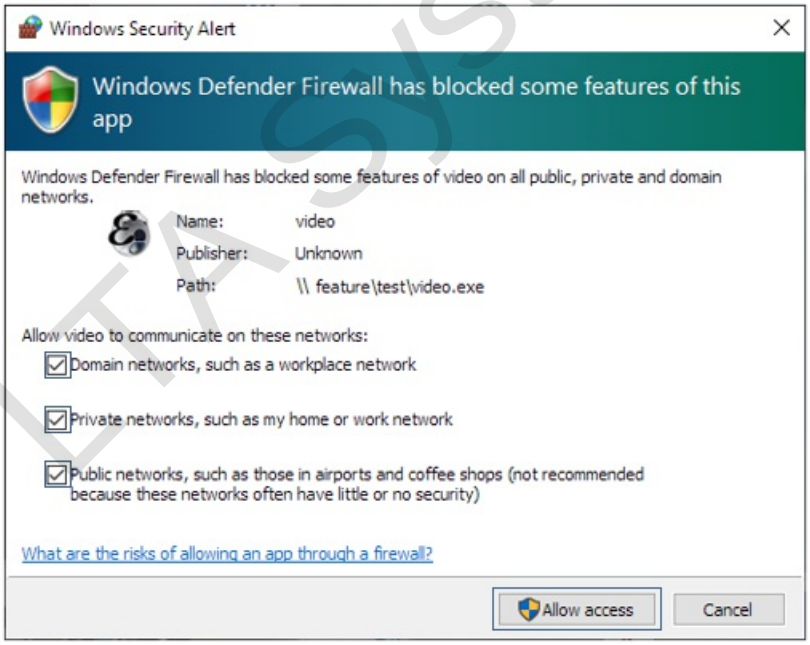
5.1 Installation

To start using the Epsilon Control application:

1. **Copy Epsilon Control application files to computer.**
   The files are on the USB stick provided with Epsilon hardware. To ensure that the application works correctly, create dedicated folder under C:/ (e.g. C:/Epsilon/, do not run the application from C:/Program Files/...
2. **Run Epsilon.exe.**
   Depending on computer security settings, security warning might pop-up – if this happens, just accept and Run the application. Also, it will ask access to networks – it is important to select both Private and Public networks to ensure that app works correctly.



Security warning pop-up

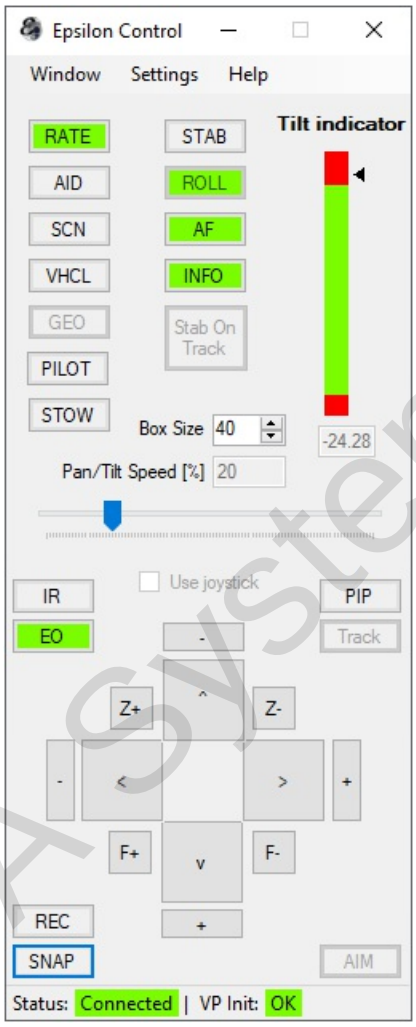


Example of private and public access

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5.2 General

Epsilon gyro-stabilized gimbals can be controlled using Epsilon Control application. The Application uses RS232 communication interface and provides Epsilon operation control such as Pan/Tilt control, zoom, focus, tracking, SD recording as well as basic configuration such as Digital video IP settings, frame size, bitrate and other. The Epsilon Control application runs on Windows 7, 10.



Epsilon Control application

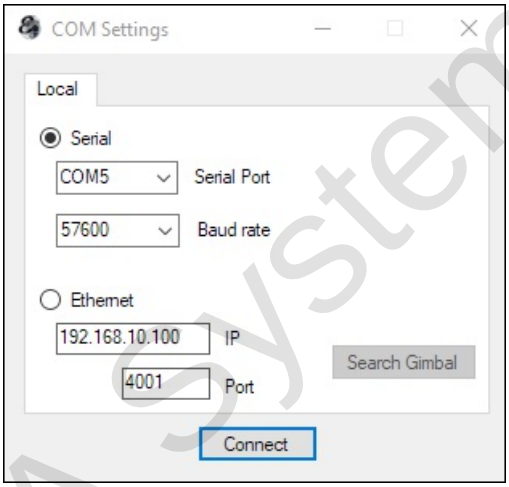
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5.3 PC Connection

There are two ways to connect to Epsilon from the computer for control it:

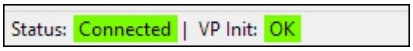
- RS232 serial port with baud rate 57600. The USB to RS232 converter can be also used in case PC does not have direct RS232 interface. In this case, the converter provides virtual COM port interface, which is used in the same way.
- Ethernet connection - the PC must be set to be in the same subnet with Epsilon. For example, Epsilon default IP is set to 192.168.10.100 and PC must have the same subnet, different last number for IP, for example, 192.168.10.11. The user can also select any other IP settings as well as or to set DHCP mode to use router. To set up your computer's network, refer to section **5.4 - PC Network configuration**.

Open **Window > Connect** to establish a connection to the Epsilon. The COM Settings window will appear. Select either serial or Ethernet communication (by choosing the appropriate checkbox) and press **Connect** button. For Serial connection, correct Port and Baud rate must be chosen. In the case of Ethernet communication, enter the IP address and port to which you want to connect to on the Epsilon gimbal.



COM Settings window

After communication is established, the Status indicator on the bottom will indicate green status. This means that Epsilon is responding to the packets sent from the application.

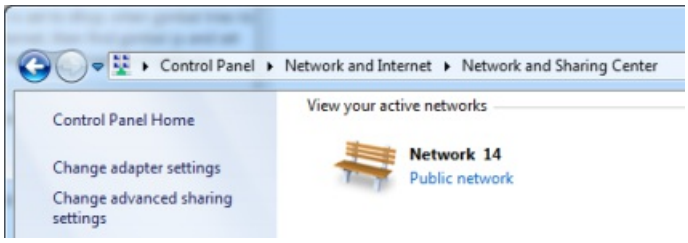


Communication status

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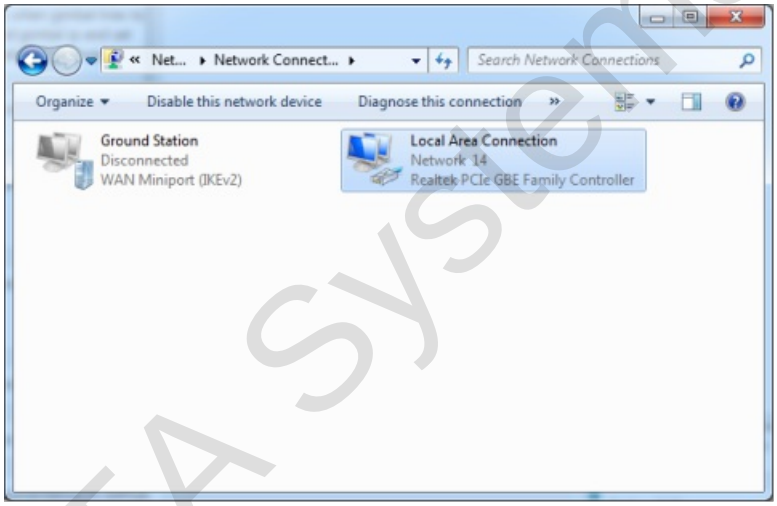
5.4 PC Network Configuration

To set an Ethernet communication between compuer and Epsilon unit, the computer network adapter must be configured. Navigate to Network and Sharing Center in the Windows Control Panel and open Change Adapter Settings.



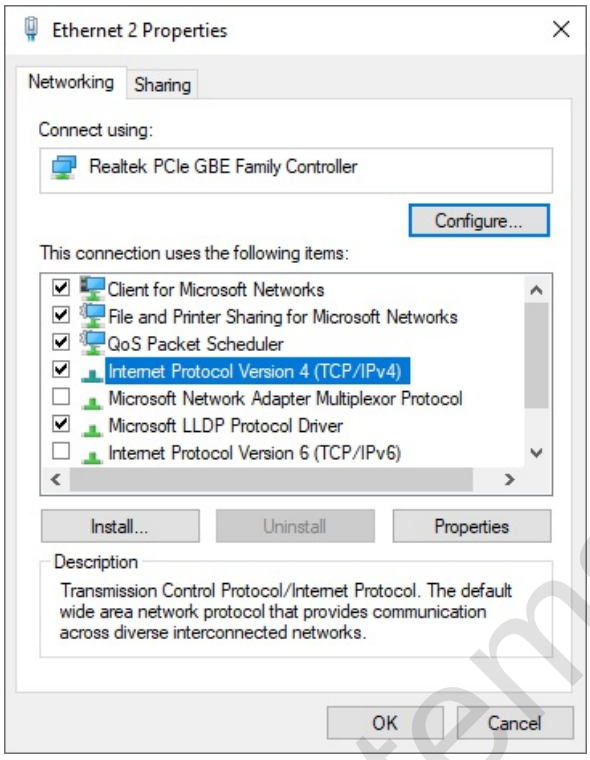
Network and Sharing Center

In the adapter settings window select a Network Adapter, which is connected to the Epsilon via the Ethernet interface. Right mouse click to select current Network Adapter Properties. The adapter properties window will appear. Double-click on the Internet Protocol Version 4 (TCP/IPv4) and enter your IP settings as shown. The PC must be configured to be in the same subnet as Epsilon.

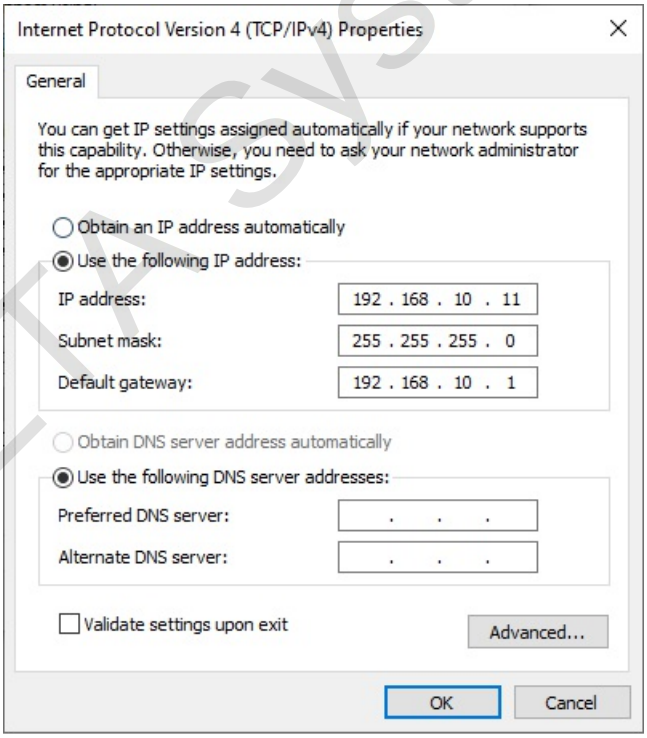


Network Adapters Window

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Network Adapter Properties



Network Adapter IP settings

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5.5 System Information

Epsilon unit system information and available feature list can be found in Help menu. About window shows information such as system versions, serial number, configuration and available Epsilon features.



About window

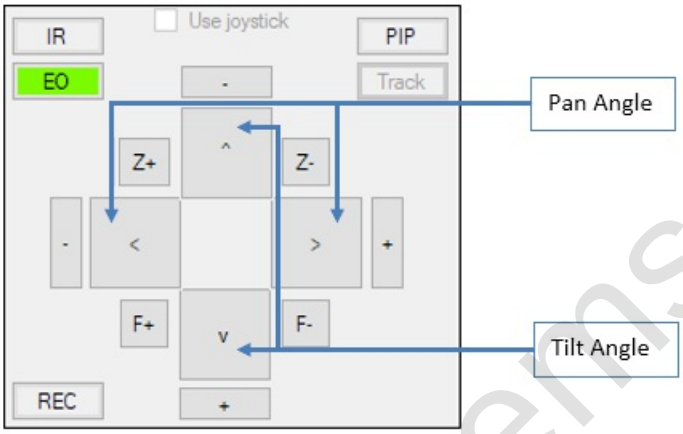
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5.6 Main Window

The main application window has basic Epsilon operation functions and control such as Pan / Tilt control, zoom, focus, tilt angle indication, and others. Other windows can be open from the window or settings menus.

5.6.1 Pan / Tilt Control

Pan and Tilt angles are controlled by rate buttons.



Pan / Tilt rate control

5.6.2 Speed Limit

Pan/Tilt rate control can be limited by the Pan/Tilt speed limit slider. The default value is 20 and the maximal value is 100.



Pan/Tilt speed limit

5.6.3 Trims

In case of small drifts on the Pan or Tilt axis, the trim can be applied to the controller. Trim buttons are placed near the Pan and Tilt rate buttons respectively and named as + and – .

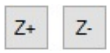


Trim adjustment buttons for pan and tilt axis

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5.6.4 Zoom Control

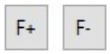
Zoom can be controlled by the zoom control buttons. For the EO camera, it will first zoom optically to the limit and then continue to zoom digitally. For the IR camera if optical zoom is available (140Z and 140Z G2) it will first zoom optically and then continue to zoom digitally. If no optical IR zoom is available (140 and 140LC) it will only zoom digitally.



Zoom control buttons

5.6.5 Focus Control

The Focus can be controlled manually by focus control buttons. IR camera focus adjustment is valid only for the Epsilon 140Z.



Focus control buttons

5.6.6 Box size

In SCN and VHCL mode size of the box can be configured to suit the object that is tracked.



Box Size

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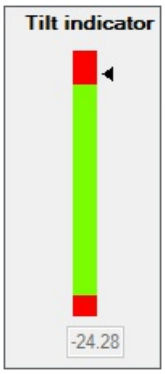
5.6.7 Basic Function Buttons

The Epsilon Control main window has basic function buttons.

RATE	Pan/Tilt manual steering mode. Uses Pan/Tilt control buttons or joystick.
AID	Pan/Tilt manual steering mode. This mode sets Pan/Tilt angular acceleration.
SCN	Scene tracking mode. This mode is used to track non-moving object.
VHCL	Vehicle tracking mode. This mode is used to track moving object such as car, boat, human and other.
GEO	GEO lock mode. Refer to Epsilon GEO-location User Manual for more detail.
PILOT	User configured custom Tilt/Pan angle for holding gimball position
STOW	Gimball camera is positioned and held in the upper position not to damage during landing
STAB	Onboard image stabilization. Indicates green light when stabilization is ON.
ROLL	Onboard image rotational stabilization. The rotation limit is configured in Common settings (section 4.5)
AF	Focus mode for daylight camera. Indicates green light when Focus is set to auto mode.
INFO	Onscreen information such as Tilt angle position, FOV, focus, and others are displayed on the video screen. Right clicking this button allows configuring what is displayed on the screen. For more information see **5.16.2 - Video overlay**
Stab On Track	Digital stabilizing that centers active track.
IR	infrared camera. Indicates green when Infrared camera is selected
EO	Daylight camera. Indicates green when daylight camera is selected.
REC	Video recording on onboard SD card. Indicate red light when recording is ON.
SNAP	Take a snapshot. Snapshots are saved on the SD card.
PIP	Picture in Picture feature. One camera (which selected EO or IR) is displayed on the "Video windows" at the same time as the second camera is displayed in inset windows, located right bottom corner.
TRACK	Track object selected in Picture in Picture feature.
AIM	laser Pointer. Caution must be taken when using a laser device. Refer to laser safety.

5.6.8 Tilt Indication

Tilt angle indication shows real Tilt angle in degrees referenced to the gimbal base. This indicator helps to control optimal tilt angle for Epsilon. The red upper and lower zones indicate the critical Tilt angle. In this case Tilt stabilization performance drops because of tilt physical limits. The green zone is optimal Tilt angle.



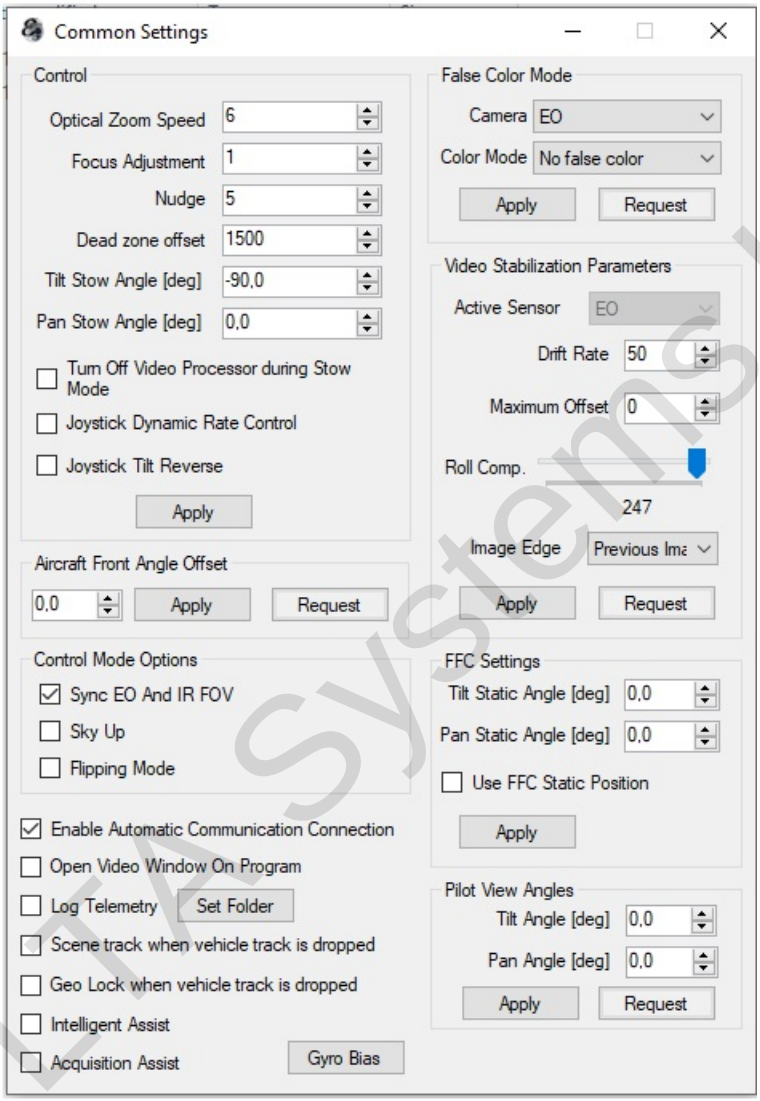
Tilt angle indication

Note that the optimal tilt angle is recommended to be not more than 60 degrees. The stabilization performance drops dramatically for tilt angles higher than 60 degrees.

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5.7 Common Settings

Common settings are located under Window menu on the main application window. Common settings window combines Epsilon common system parameters and functions such as SD recording, general control parameters, false color, KLV, video stabilization and gyro bias calibration function.



Common settings window

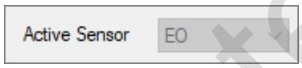
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5.7.1 Control Settings

Optical Zoom Speed	Optical zoom speed limit. The speed range is from 1 to 7. Valid for the daylight camera only. IR camera has a fixed continuous zoom speed (Epsilon 140Z/140Z G2).
Focus Adjustment	The focus adjustment step. The range is from 1 to 255.
Nudge	Nudge adjustment step in pixels.
Dead zone offset	Joystick Pan/Tilt control dead zone. Default dead zone is 1500.
Pan/Tilt Stow Angles	The Stow Angles is the position where Epsilon Gimbal should be left when it is not observing, or planned landing, to avoid camera damages during landing. Stow Angle is set by the operator.
Processor Power Off During STOW	Removes power from a Gimbal's processor in a controlled way. After processor power off power consumption will be lowered, although Pan/Tilt motors will retain power and will hold Gimbal in Stow Angles. (After STOW mode is activated, to restore gimbal processor power, need to make power reset).
Joystick Dynamic Rate	Set joystick Pan/Tilt control to logarithmical scale.
Joystick Tilt Reverse	Change direction of Tilt axis.

5.7.2 Video Stabilization Parameters

Video Stabilization parameters can be set for each sensor. Active sensor is shown in text box.



Example of currently displayed active sensor

Drift Rate	Image re-centering rate by onboard processing. Large values lead to rapid re-centering and eliminate only highest frequencies. Small values lead to slow re-centering, keeping scene nearly still. The range is 0-255. The default value for the Epsilon is 11.
Maximum Offset	An image stabilization offset (translation limit) in pixels. The default value for EO camera is 100 and the value for IR is 30.
Roll compensation	Maximum stabilization angle in degrees for cameras roll motion.

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5.7.3 False Color

False color can be applied to original gray IR image to make it more comprehensible. False color allows recognizing more details in the image. Typically, the human eye can only distinguish about 16 shades of gray from one another.

Available false color palletes:

Pallette Name	Color range ← Cold Hot →	Pallette Name	Color range ← Cold Hot →
White Hot		Black Hot	
Rainbow		Rainbow Inverted	
IRON		IRON Inverted	
Hot/Cold		Hot/Cold Inverted	
Jet		Jet Inverted	
Hot		Hot Inverted	
470CLR_S		470CLR_S Inverted	
Color 1		Color 1 Inverted	
Color 3		Color 3 Inverted	
Color 2		Color 2 Inverted	
HotIron		HotIron Inverted	
HSV		HSV Inverted	
IceFire		IceFire Inverted	
Iron256		Iron256 Inverted	
Rain256		Rain256 Inverted	
Xvolocano		Xvolocano Inverted	
IDDef		IDDef Inverted	
Red		Red Inverted	
Green		Green Inverted	
Blue		Blue Inverted	

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5.7.4 Additional Checkboxes

Open Video Window On Program	Video Window opens, when software is launched (If checkbox checked it is auto applied).
Log Telemetry	Epsilon Gimbal starts to log telemetry (If checkbox checked it is auto-applied).
Scene Track When Vehicle Track is Dropped	During Vehicle Tracking, if the target is lost - VHCL switches to SCN (If checkbox checked it is auto-applied).
Geo Lock When Vehicle Track is Dropped	During Vehicle Tracking, if the target is lost - VHCL switches to GEO (If checkbox checked it is auto-applied).
Enable Automatic Communication Connection	Epsilon Gimbal connects to the last connected Com port settings.
Intelligent Assist	Off (default), on – assists initialization of the track box size and location
Acquisition assist	Off (default), on – Requires Intelligent Assist to be enabled. The tracker will look for turning tracks and if detected will attempt reacquisition. The new track will be followed for a period of time before replacing the current track.

5.7.5 Gyro Bias

Gyro Bias	Gyro drift calibration. The gimbal must not move during the calibration. This function calibrates gyro drift. Pan and Tilt motors are off for 5 seconds.

5.7.6 Flat Field Correction

Flat Field Correction	Storing the supplemental flat field correction values to apply to thermal image data of the infrared camera to adjust for non-uniformities associated with the first optical path.
Pan/Tilt FFC Angles	Position, where Flat Field Correction is to be made in case the same static location, is used every time.

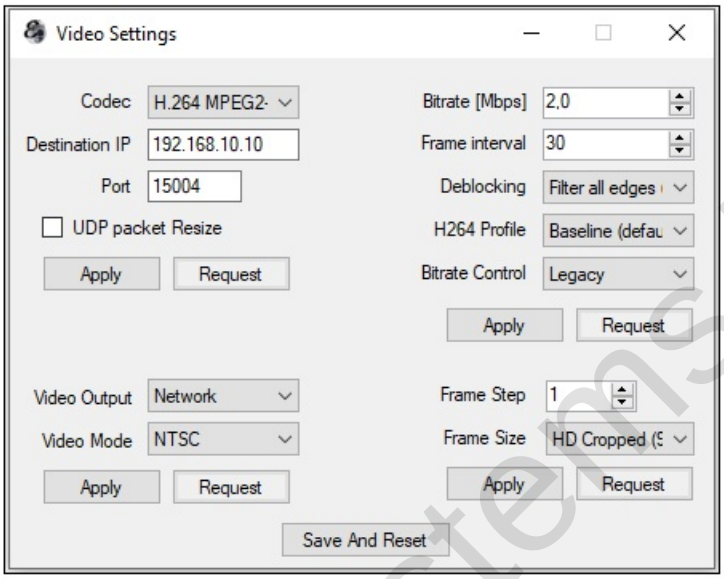
5.7.7 Control Mode Options

Sync EO and IR FOV	both FOV 's are synchronized, between EO and IR camera. Currently, this feature is supported by 140 and 140Z.
Sky Up	For gimbals that are mounted on the nose of UAV. In this mode image are rotated according to camera viewing angle to make it always correct. Currently, this feature is supported by 140LC, 140Z G2 and 140Z.
Flipping Mode	In this mode selecting between infrared or daylight camera makes gimbal flip automatically, making the active camera always to be in the lower position. Currently, this feature is supported by 140LC, 140Z G2, and 140Z.

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5.8 Video Settings

Video settings are located under the Window menu. After changing any of the settings "Apply" button must be pressed to send these settings to the gimbal. They will be valid only for the current session while the gimbal is powered on, only exception is "Frame Step" it will be saved permanently after pressing the "Apply" button. If other settings must be permanently set then after pressing the "Apply" button user must also press the "Save and reset" button.



Video settings window

Codec The following codecs can be selected:

- RTP MJPEG
- H.264 MPEG2-TS
- H.264 (HD) - (Default)
- RTP H.264 (No metadata)
- RTP MPEG2-TS H.264 (+ metadata)
- None

Destination IP The destination IP address. This address must be set the same as PC IP address which is used for digital video receive. For multicast stream use 224.10.10.10.

Port The default port is 15004. Do not change the port number if using Epsilon Integrated Video Viewer.

Video Output The default is Network for digital video. The analog or network and analog together can be selected.

Video mode Analog video mode. PAL or NTCS can be selected for analog video.

Bitrate Measured in Mbps.

I-Frame Key frame interval. The default value is 30.

H264 profile With profiles, the encoder and decoder agree on a feature set that they can both handle.

Bitrate Control Provide the ability to increase quality (relative to legacy rate controls) while maintaining the bitrate constant

Frame Step Frame interval to be displayed. The default value is 1 which corresponds to 30 frame/s.

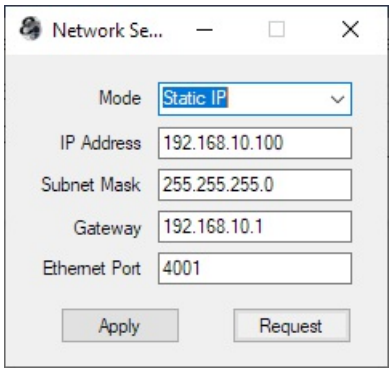
Frame size Displayed frame size. SD (640x480), HD Cropped (960x720), and HD (1280x720) is supported for all Epsilon 140 series. Full HD (1920x1080) is available only on 140Z G2.

Save And Reset To save video settings. This operation requires reset for video processor unit. It will take around 30 seconds. The video will be lost while restarting.

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5.9 Network Settings

Network settings are located under the Window menu on the main application window.



Ethernet settings window

Mode	The default mode is Static IP. DCHP can also be used.
IP Address	The IP address of the onboard Image processing unit. The default IP is set to 192.168.10.100
Subnet Mask	The default mask is set to 255.255.255.0
Gateway	The default gateway is set 192.168.10.1
Ethernet Port	Port used for gimbal cotrol

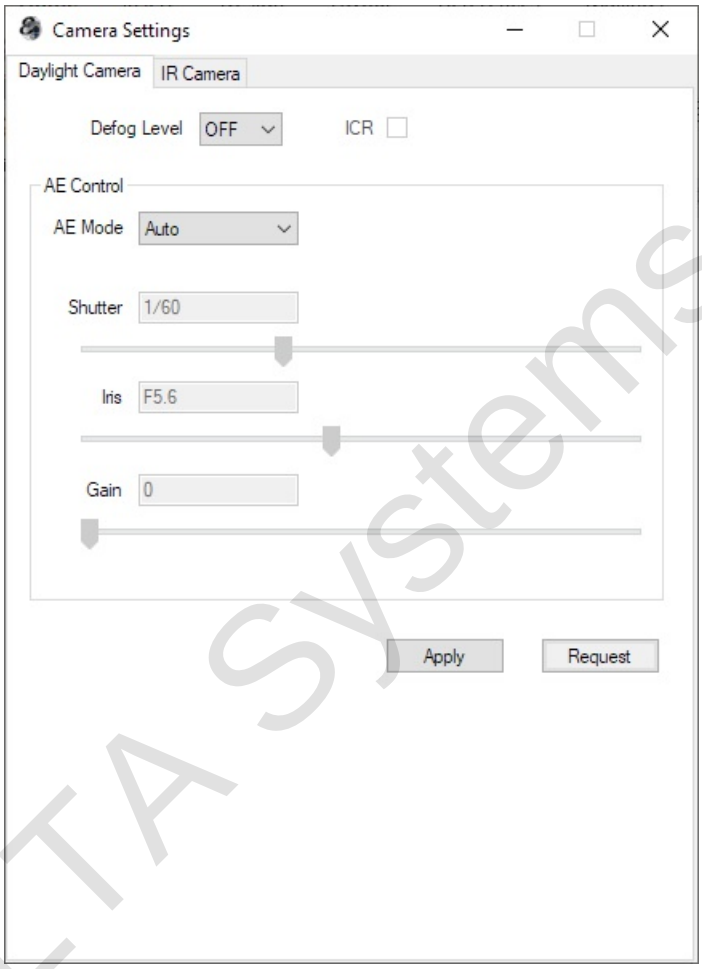
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5.10 Camera Settings

Camera settings are located under Window menu on the Epsilon main application window. Camera settings window includes EO and IR settings tabs.

5.10.1 EO Sensor

EO Camera settings window:



Camera settings window

Defog Level – improves image sharpness in fog or clouds conditions by the dynamic contrast control. There are three defog levels. The Level 1 has the weakest effect and Level 3 has the strongest effect. Defog is reset to zero after power cycle.

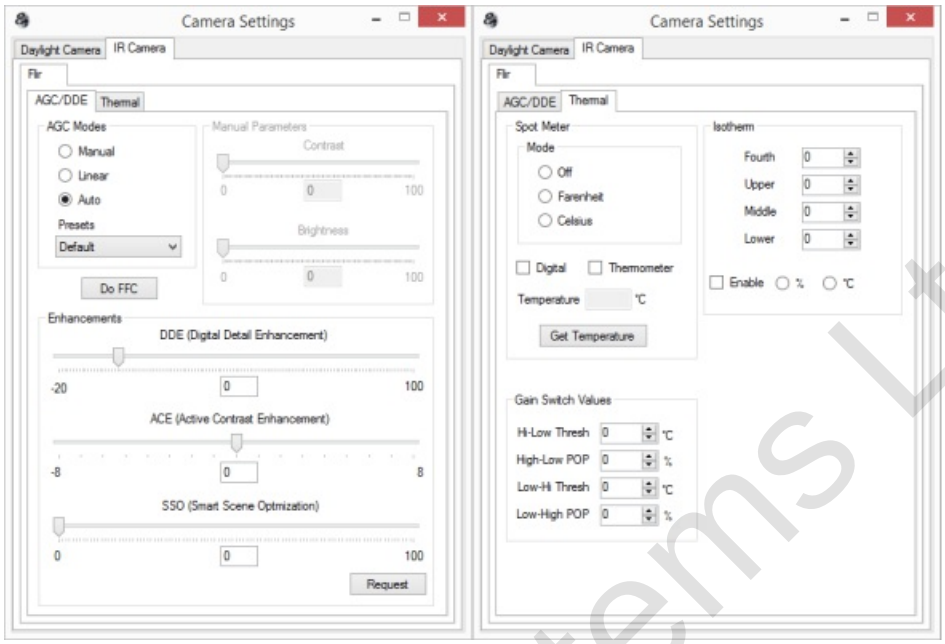
AE (auto exposure) control – the default AE Mode value is set to auto. With Shutter Priority AE, once the user sets the shutter speed the camera will automatically sets the most appropriate iris (aperture) and gain values. With the Iris Priority AE, once the user sets the desired iris, the camera automatically sets the appropriate shutter speed and gain. The full manual mode is also available to set shutter speed, iris and gain using corresponding sliders.

ICR (IR cut Filter Removable) – the command removes infrared filter, increasing the sensor sensitivity. The image becomes black and white. This mode is useful in sunrise or sunset time when there is not enough light for auto AE mode. ICR is disabled in auto AE mode.

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5.10.2 IR Sensor

IR settings window is shown below:



IR sensor settings

AGC (Automatic Gain Control) modes – IR sensor (FLIR TAU2) supports three AGC modes: automatic, manual and linear (linear histogram). Automatic mode is most sophisticated for most situations. In this mode, image contrast and brightness are optimized automatically. In the manual mode, image contrast and brightness are controlled manually using corresponding sliders (on the right side). In linear mode, mage contrast and brightness are optimized automatically on scene using a linear transfer function.

Do FFC (Flat Field Correction) – FFC is used to correct non-uniformity of the IR image. Camera should be pointed on the uniform area with max zoom (140Z) before pressing this button.

DDE (Digital Detail Enhancement) – DDE is typically used in the automatic AGC mode where only one threshold is used to determine the enhancement on a per scene basis. DDE range from -20 to -1 soften the image, zero is off and 1 to 100 sharpens the image and helps accent edges and other high frequency spatial details.

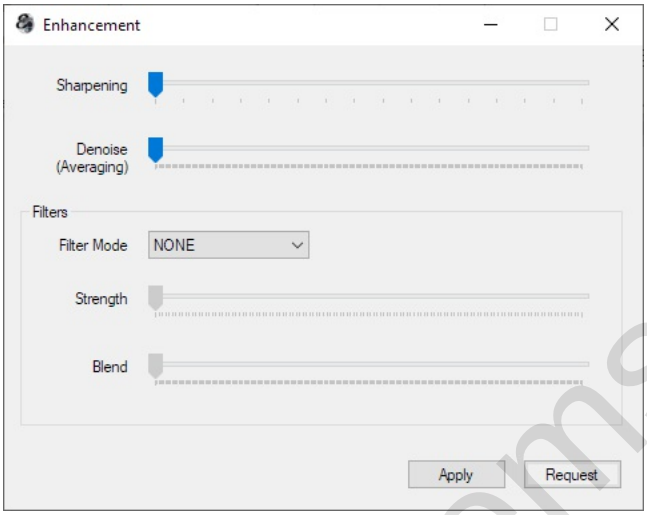
ACE (Active Contrast Enhancement) – The feature provides a contrast adjustment dependent on the relative scene temperature. ACE thresholds greater than 0 impart more contrast to hotter scene content and decrease contrast for colder scene content (e.g. sky or ocean). ACE threshold less than 0 do the opposite by decreasing contrast for hotter scene content and leaving more of the gray-scale shades to represent the colder scene content. ACE can range from -8 to 8 where -8 decreases contrast in darker regions while +8 increases contrast in darker regions.

SSO (Smart Scene Optimization) – SSO value defines the percentage of the histogram that will be allotted a linear mapping. Enabling this feature facilitates the avoidance of irradiance level compression, which is specifically important for bi-modal scenes. With SSO enabled, the radiometric aspects of an image are better preserved (i.e. the difference in gray shades between two objects is more representative of the difference in temperature). SSO helps to set how linear the HEQ and information based AGC algorithms can be to provide the highest amount of perceived contrast in every scene. SSO is a percent and can range from 0 to 100%.

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5.11 Video Enhancement

Video Enhancement is optional video processing feature, which provides Contrast Limited Adaptive Histogram Equalization (CLAHE) and Local Area Processing (LAP) functions to improve output video. The Video Enhancement window is located under Settings menu on the main application window.



Video Enhancement Window

Filter Mode – LAP (Local Area Processing) is a large linear filter, which emphasizes small features and dampens slightly large features. It is like edge sharpening. CLAHE (contrast limited adaptive histogram equalization) increases the contrast of low-contrast regions of an image without affecting the contrast of high-contrast regions.

Sharpening – Applies a small linear filter to the image that enhances the visibility of edges and small details in an image. Since it also tends to emphasize sensor noise and so is often used with Denoising.

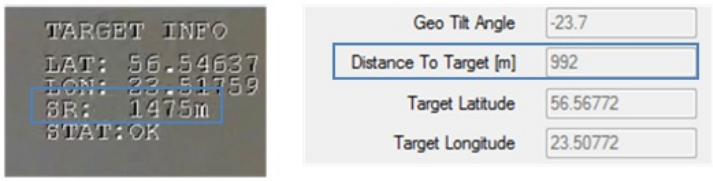
Blend – Blending filtered video with raw video. If blend level is zero, the applied filter does not affect the output image. For 50% blend level, filtered image is blended with the raw image and respectively 100% corresponds to display filtered image only.

Strength – Applied filter strength.

Denoise – Performs frame averaging to reduce image noise. However, if there is frame-to-frame motion the output image exhibits motion-induced blur.

5.12 Laser Range Finder

The Laser Range Finder (LRF) is typically used for distance measuring of static and moving objects. LRF is based on laser diode technology with wavelength of 1.55 µm, which is not visible to IR sensor. The laser is Class 1 type, which is eye-safe.



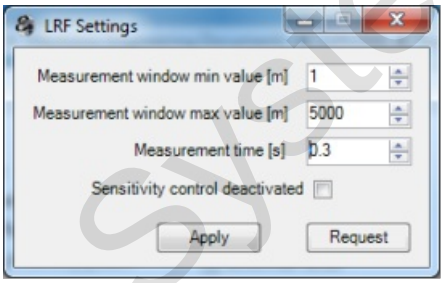
LRF Distance on Video overlay (left) and in Telemetry Window (right)

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The maximum range of the Laser Range Finder depends on few parameters like target size, Albedo, vibration, visibility and measurement time. Typical measurement range is shown in Table 2.

Typical LRF range	
Condition	**Range**
Man Size Target (0.75 x 0.75m, Albedo 30%, 10km Visibility)	~ 2000 meters
NATO Standard Target (2.3 x 2.3m, Albedo 30%, 10km Visibility)	~ 3000 meters
Extended Target (6 x 6m, Albedo 50%, 23km Visibility)	~ 4500 meters

LRF settings window is located under Settings menu on the Main Epsilon application window. Measurement window can be set as well as measurement time. The default measurement time is 0.5 seconds. For static platforms in low vibration environment, the measurement time can be higher (up to 3 seconds), which results more stable and accurate distance measurement. However, going lower than 0.5 second will decrease LRF maximum range.



LRF Settings Window

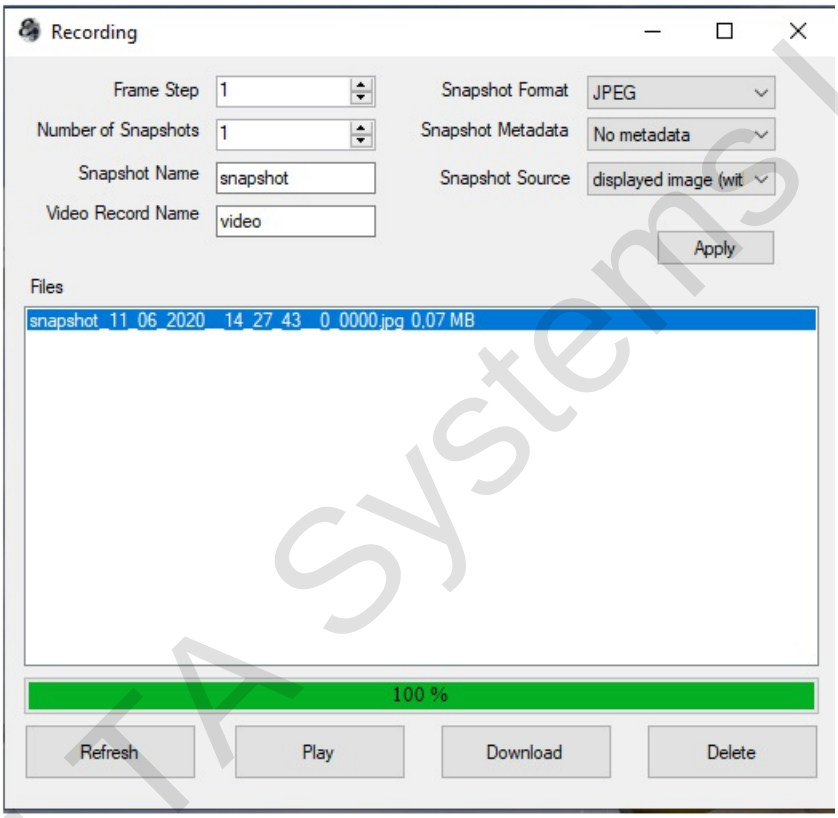
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5.13 GEO Settings

Please refer to Epsilon GEO-location User Manual.

5.14 Recording

All recorded video files and snapshots are stored in build-in SD card. Files can be downloaded or deleted from the SD files browser, which is located under Window menu. Press Refresh button to Read or Update SD card files information. Video and snapshots files are displayed with defined names (refer to section 5.6) with added date and time information. The Video file is recorded as TS type file, which can be played with VLC player.

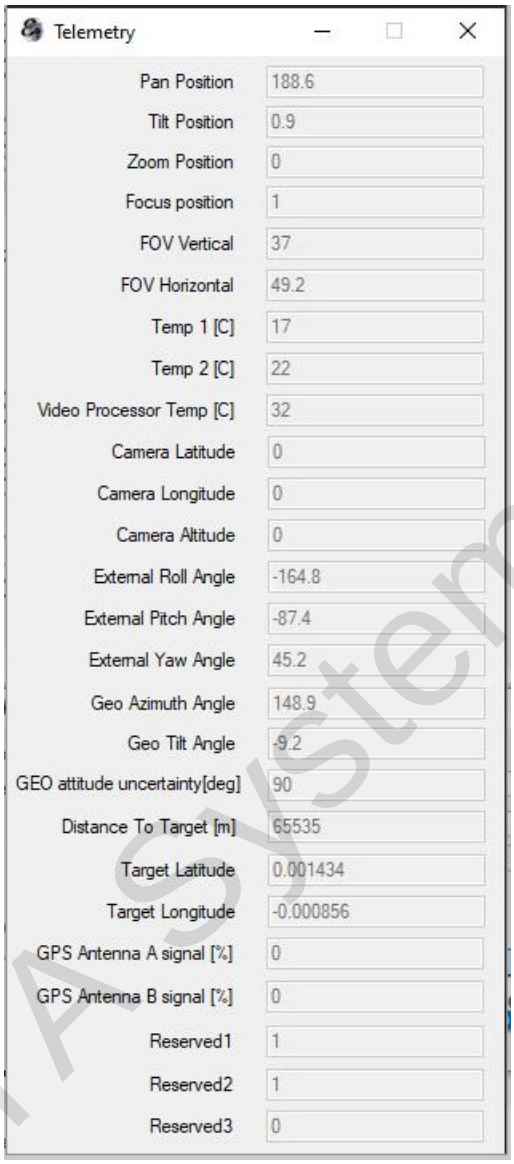


SD files browser

Frame step	Interval of frames between taking snapshots
Number of snapshots	Number of snapshots to be taken on the one button click. The Frame Step parameter is used when taking more than one snapshot
Snapshot Name	Snapshot file name to be stored on SD card
Video Record Name	Video file name to be stored on SD card
Snapshot Format	Setting for snapshot format when using SNAP
Snapshot Metadata	Setting for snapshot metadata when using SNAP
Snapshot Source	Source where snapshot is taken from

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5.15 Telemetry



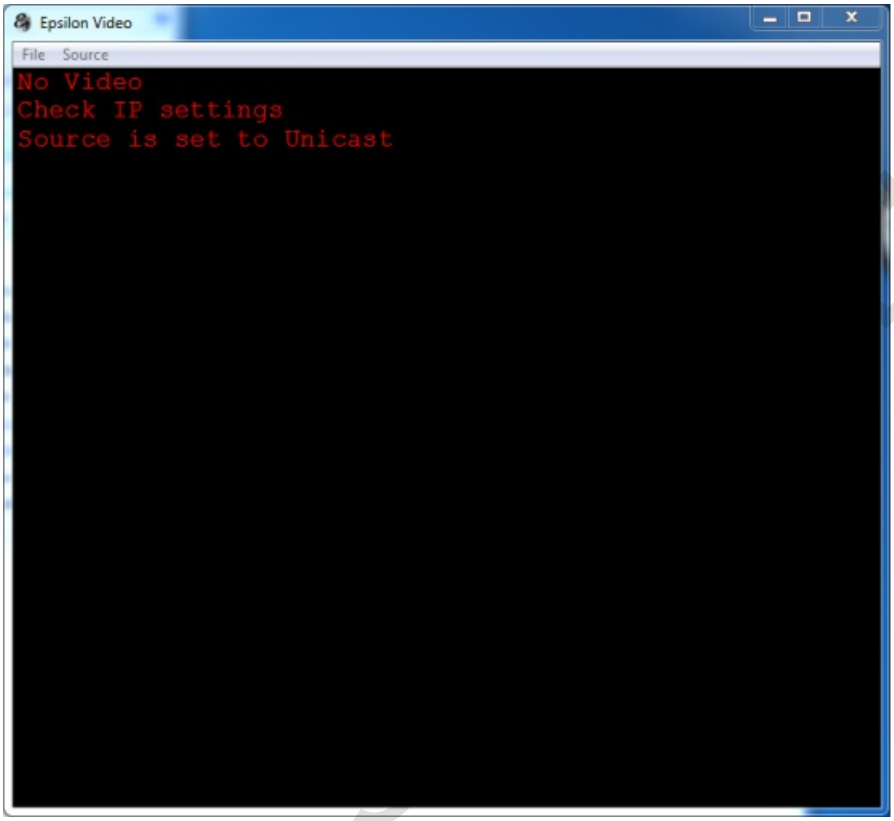
Telemetry window

Telemetry window allows seeing precise information about current parameters of gimbal, such as position of pan and tilt, vertical and horizontal FOV , temperature inside the gimbal, external angles of gimbal, position of gimbal, and signal strength of GPS.

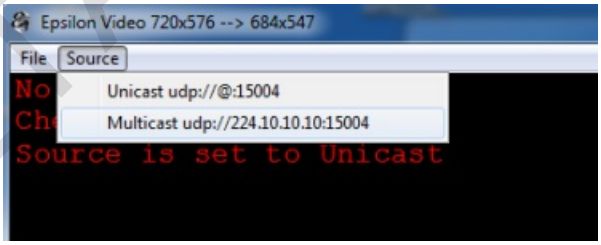
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5.16 Digital Video

The integrated video viewer can be used to view video stream from the Epsilon. The Epsilon video viewer can be opened by clicking on Video from the Window menu. The viewer uses the default port 15004 for video stream. The destination IP address must be configured in the Video settings (refer to section **5.8 - Video Settings**).



If Epsilon is set to stream Multicast video then the Source also must be set for multicast. To set multicast video, click on Multicast UDP under the Source menu.

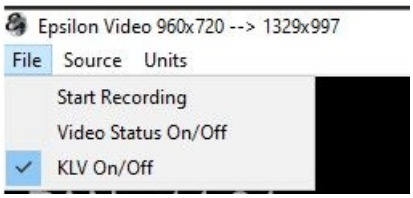


Source selection

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5.16.1 Epsilon Video KLV toggle

Epsilon video player allows to toggle KLV data to be either displayed or not, it is available under "File" dropdown menu.



KLV toggle option for video player

5.16.2 Video overlay

Additional information can be enabled or disabled by right-clicking the "INFO" button on the main Epsilon window. The summary of where each info is displayed is shown in the illustration below.

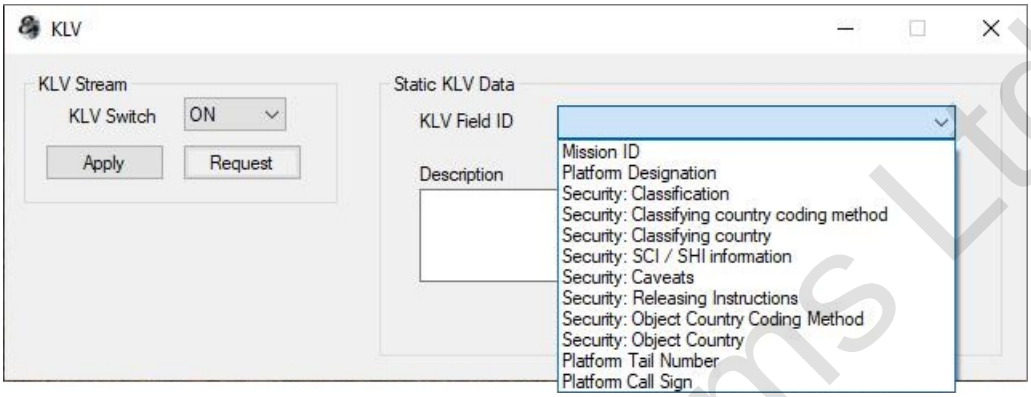


Video overlay positions for each item under the INFO button

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5.17 KLV

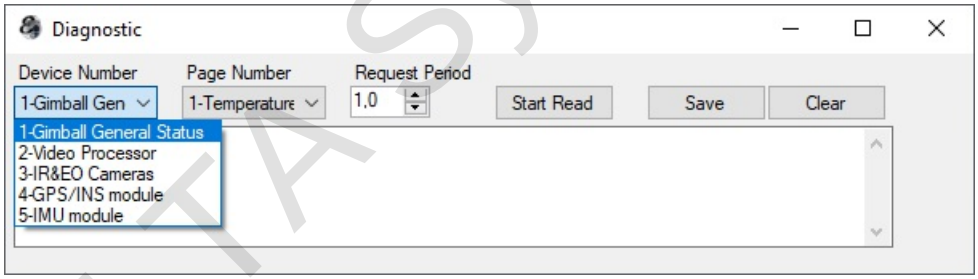
KLV metadata is generated in accordance with MISB standards 0102.10, 0601.7, 0603.2, 0604.3, and 0903.3. KLV settings allow to turn KLV stream on or off by selecting the required state and pressing "Apply". ID fields can be filled in with the required description to be displayed in the KLV stream. ID fields must be filled in accordance with the MISB ST 0601.14 standard "Table 1: UAS Datalink Local Set". If the field ID is left with an empty description then it will be removed from the KLV stream.



KLV window

5.18 Diagnostic

Diagnostic tool for Gimbal, main information- temperature, motor current, and errors can be viewed for different parts of the Gimbal. Diagnostic is started by pressing "Start Read". Interval, when the data is received, is set by changing the "Request period" in seconds. The minimum request period is 0.1 seconds.

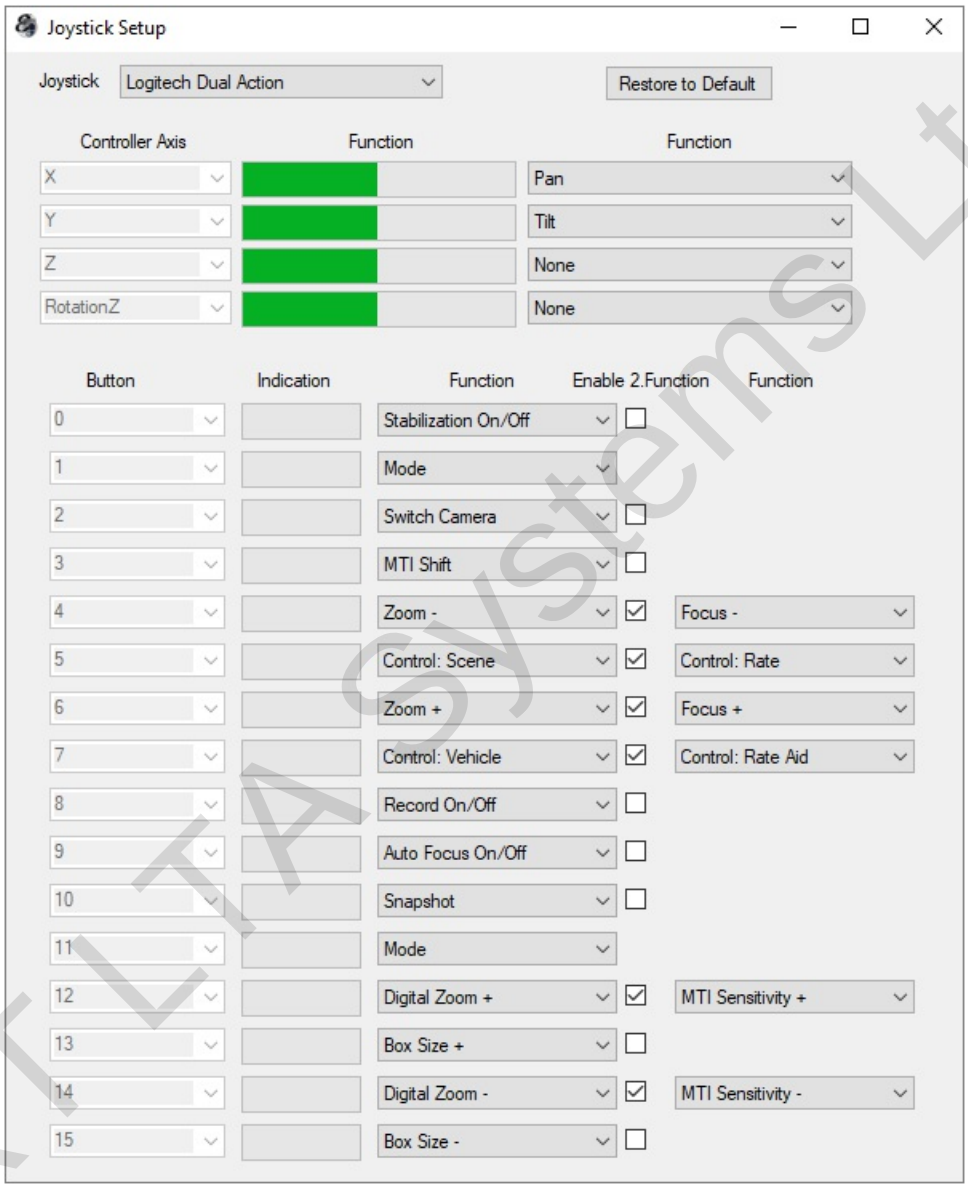


Example of diagnostic window

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5.19 Joystick Setup

Joystick Setup allows user to either restore settings to default or set custom settings for joystick buttons. Functions as well as secondary functions can be added or changed to better suit user preferences. For enabling joystick and controlling gimbal using default configuration see chapter **6-4 - Joystick Control**. In order to configure joystick buttons- when button is pressed it will light up green in Joystick Setup "Button" section, so Function can be assigned to that specific button.



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6.1 Object Tracker

The Onboard Image processing unit is equipped with an object tracking feature, which allows locking Epsilon on stationary objects such as buildings using scene mode as well as track moving objects. Objects tracking is initiated by pressing SCN or VHCL button on the Epsilon main window as well as using Joystick (refer to section 6.4). The central cross indicates the center of the tracking box for the tracked target.



Scene tracking mode

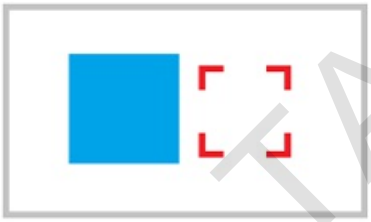


Vehicle tracking mode

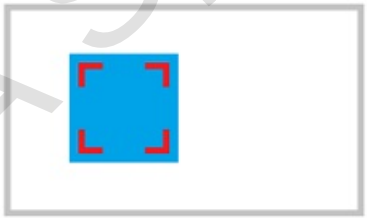
6.1.1 Selecting Target Box Location And Size

The optimal size of the tracking box should include the entire object that is to be tracked. For best performance, maximum of additional 10% region with the background is included. If it is not possible to include the full object then focus should be on features with high contrast on the object that is being tracked. For example, windows on a car.

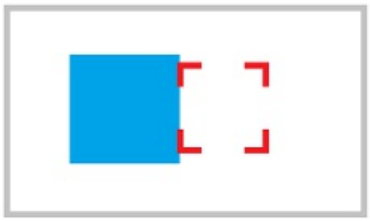
The table below shows common examples of attempting to initialize tracking. In all examples the tracked object is represented by the blue square and the tracking box is in red.



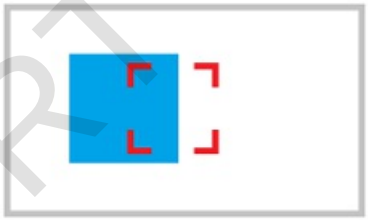
Example 1 (Bad)



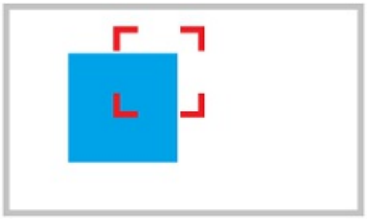
Example 2 (Bad)



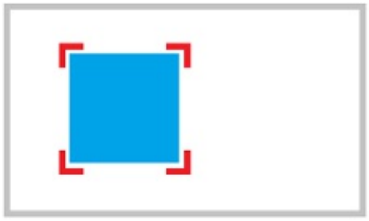
Example 3 (Bad)



Example 4 (Usable)



Example 5 (Good)



Example 6 (Best)

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Example	Performance	Comment
Example 1	None	The incorrect area is tracked instead of the required object. Tracker can either fail to initialize.
Example 2	None/Poor	As there are no visible features to work with (edges for example) tracker can either fail to initialize or drift to corners of the square.
Example 3	Poor	Only a small bit of the object is covered by the track box, In case of strong features in the background, the tracker can track those instead of the required object.
Example 4	Poor	Slightly better performance than example 3. Due to fact that the only trackable feature is a single edge the tracker can drift along the edge possibly resulting in a bit erratic track.
Example 5	Good	Better than example 4. Two edges of the object (corner) are included that the tracker can follow. Tracking will perform well if there aren't any strong features in the background.
Example 6	Best	Optimal performance is achieved when the track box includes the entire object with minimal features from the background.

Real-world vehicle tracking examples:


Tracking box doesn't cover the object


Tracking box too large


Tracking box size and location optimal

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6.2 Moving Target Indicator

Moving target indicator (MTI) is optional feature, which reduces operator workload with automatic detection of moving targets. There are two separate licenses for MTI feature: 1) baseline and 2) advanced. All available features can be found in system information section **5.5 - System information**. MTI configuration window can be found in Window menu. Moving Target Indicator feature has 5 different modes:

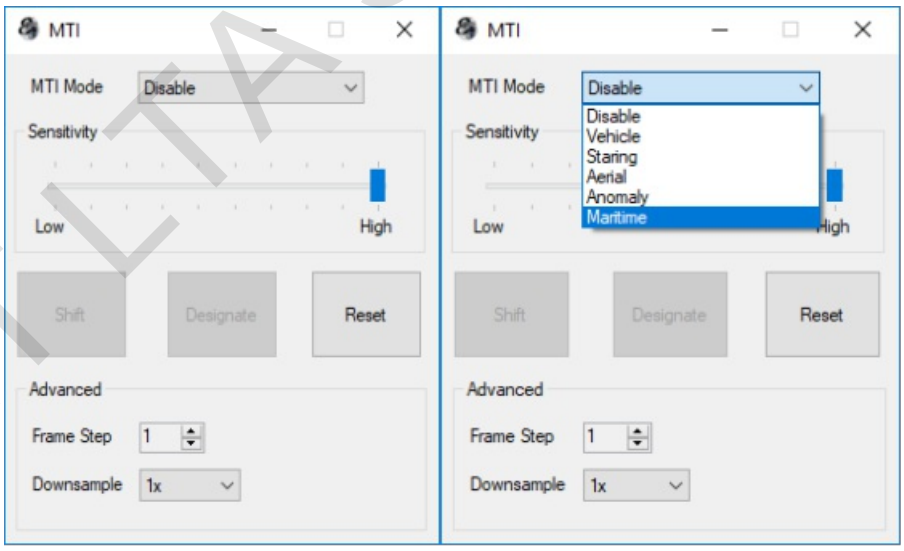
Mode	Description	License
Vehicle	Works well for finding moving objects that are typically 10 to 100 pixels long with relative constant speed. Not suitable for very small, very large, slow or erratically moving objects	Baseline
Staring	Mode is used for small, slow, fast, or varying rates objects while Epsilon is unmoved (fixed scene)	Baseline
Aerial	Works on a moving Epsilon for finding small, slow, fast, or erratically moving objects. The mode increases video processor load to handle scene motion.	Advanced
Anomaly	Works from a moving or stationary platform and finds objects that are unique anomalies or outliers in the scene based on color and/or intensity. This mode doesn't require objects to be moving to be detected.	Advanced
Maritime	Works from a moving or stationary platform and finds objects in a maritime environment such as ships or life jackets.	Advanced

Other parameters configurable in MTI window:

Sensitivity – higher sensitivity can trigger more false positives but has a higher chance of locating the target. Too low sensitivity can ignore some of the results.

Frame Step – step between frames. 1 processes every frame, 2 processes every other frame, etc. (1 to 5). Default is 1.

Downsampling – the process of reducing the sampling rate of a signal. This is done to reduce the size of the data. (0: None; 1: 2x2 downsample; 2: 4x4 downsample; 3: 8x8 downsample; 255: Auto based on frame size).



MTI window

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When vehicle mode is selected video processing unit automatically detects moving objects and indicates those with yellow box. Operator can shift between several detected objects by pressing **Shift** button in the MTI window. The green box color will shift between detected objects showing the preselected target which can be designated as primary.



Automatically detected moving object (yellow box)

Figure below shows selected target by pressing **Shift** button. When target is selected (green box) the **Designate** command will switch tracker to track the target.



Selected target by operator (green box)

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Designated target track

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Figure below shows typical aerial MTI mode. This mode helps to detect small erratically moving objects, which are difficult to find for operator. The mode does not provide automatic tracking for detected objects.



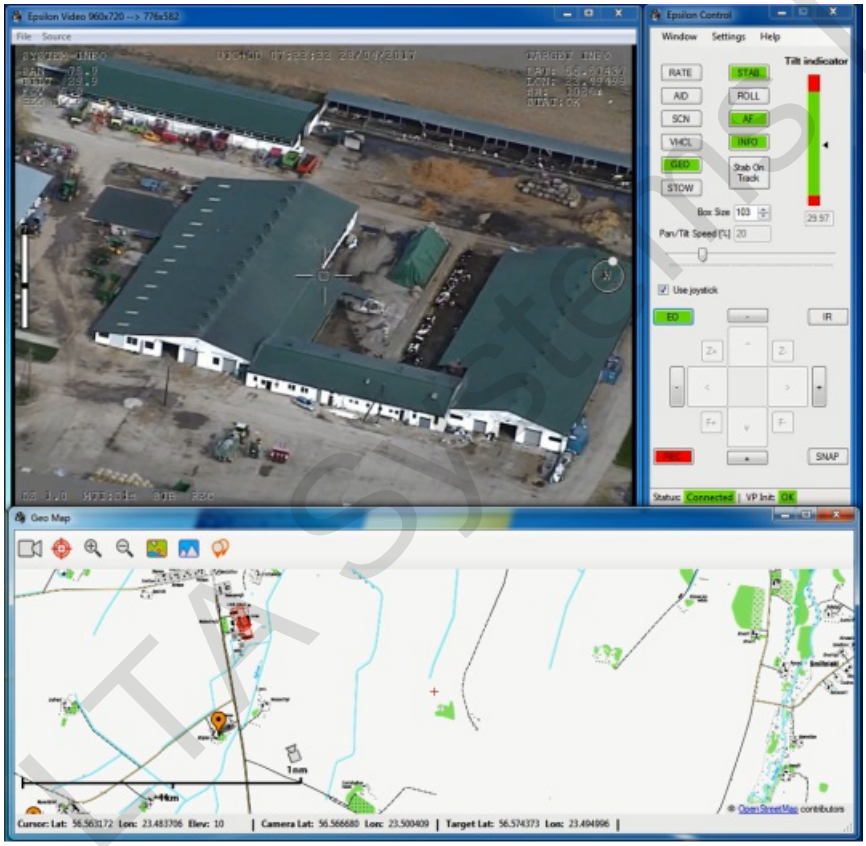
Moving human is detected by advanced MTI

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6.3 GEO Mode

Epsilon GEO Location feature allows operator to see GPS coordinates of the target at any time. At all times GPS coordinate (longitude, latitude) of the target and slant range (distance) from the target to the camera are indicated on the screen. Geo Location feature enables the following options:

a. Operator can see the GPS coordinates of the target at which the gimbal is pointing at any time;
b. Operator can see the slant range to the target at which the gimbal is pointing at any time;
c. Operator can input GPS coordinates of the object of interest and automatically steer the gimbal towards it;
d. Operator can input the list of predefined targets by entering its corresponding GPS coordinates and automatically steer the gimbal towards each one of them when necessary.



GEO mode operation

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Additional optional functionality can be enabled with GEO Moving Map Software Overlay plugin. When Epsilon GEO Moving Map is enabled, in the additional software window operator can see:

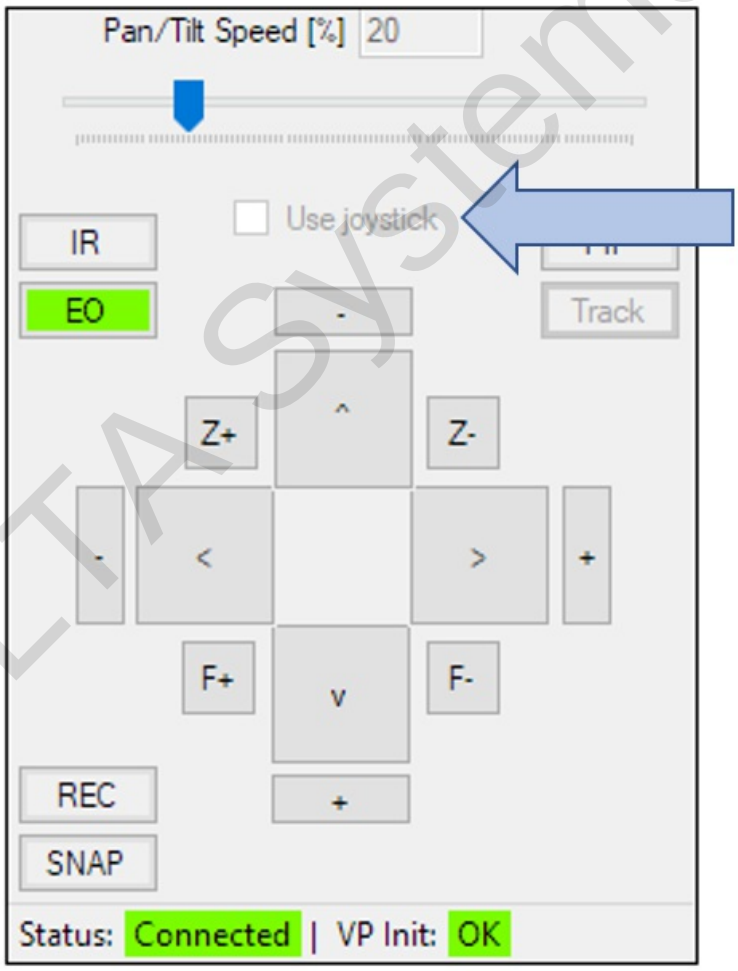
a. Exact point on the map in which gimbal is currently pointing marked with the red “target” icon;
b. The area within the current Field of View on the map marked with a red overlay;
c. Gimbal current position on the map marked by the “camera” icon;
d. Gimbal icon direction always shows the direction where the camera is actually pointing.

Operator can also select the object of interest on the map by clicking on it and utilizing “Look Here” feature.

For more details on GEO mode operation and settings, please refer to Epsilon GEO-location User Manual.

6.4 Joystick Control

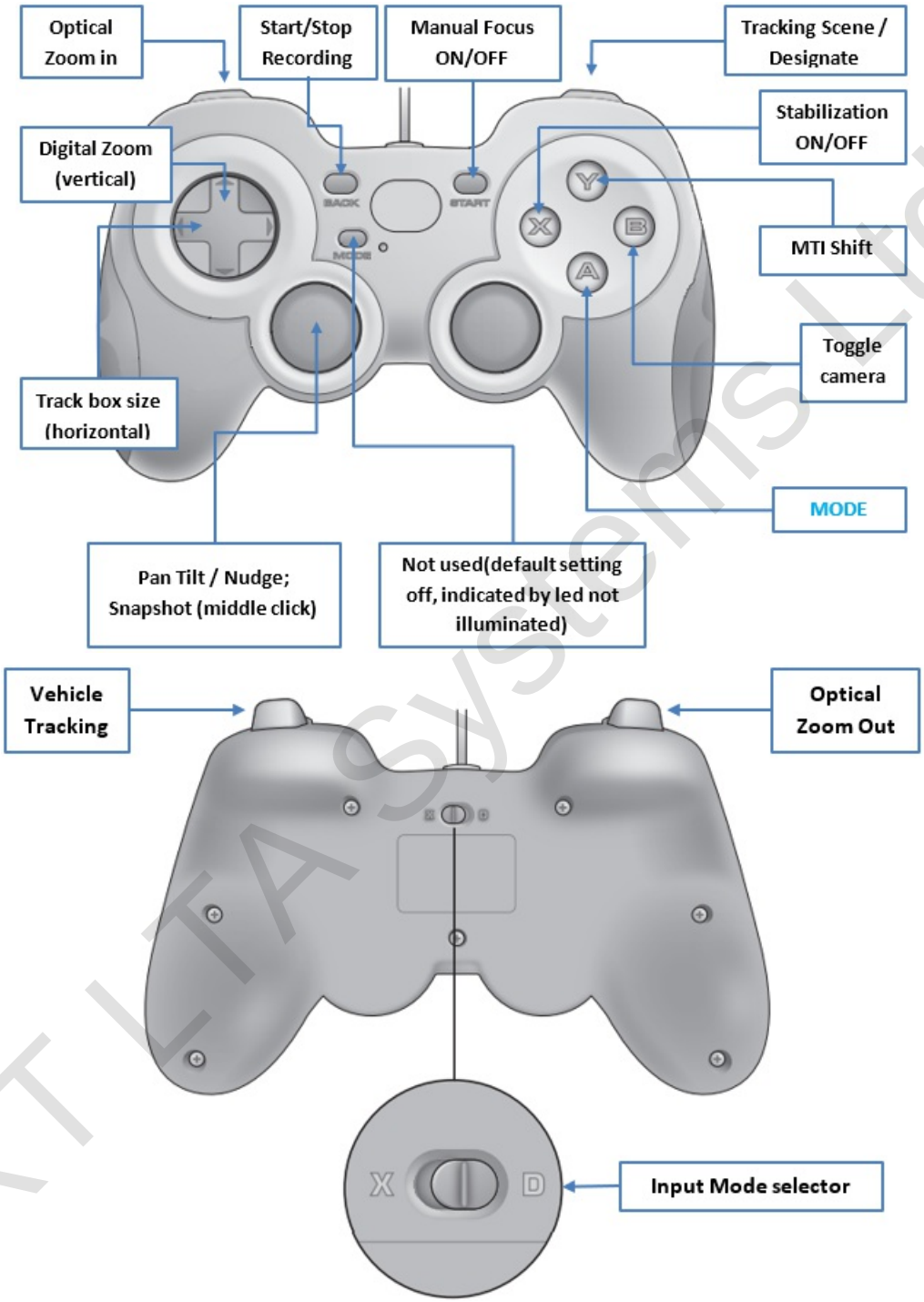
The Epsilon Control application currently supports two types of USB joysticks to operate the Epsilon: The Logitech DUAL ACTION Joystick and APEM IP Desktop joystick. To enable joystick control, connect Epsilon Control application to the Epsilon, plug USB joystick and check Use joystick option.



Enabled joystick control

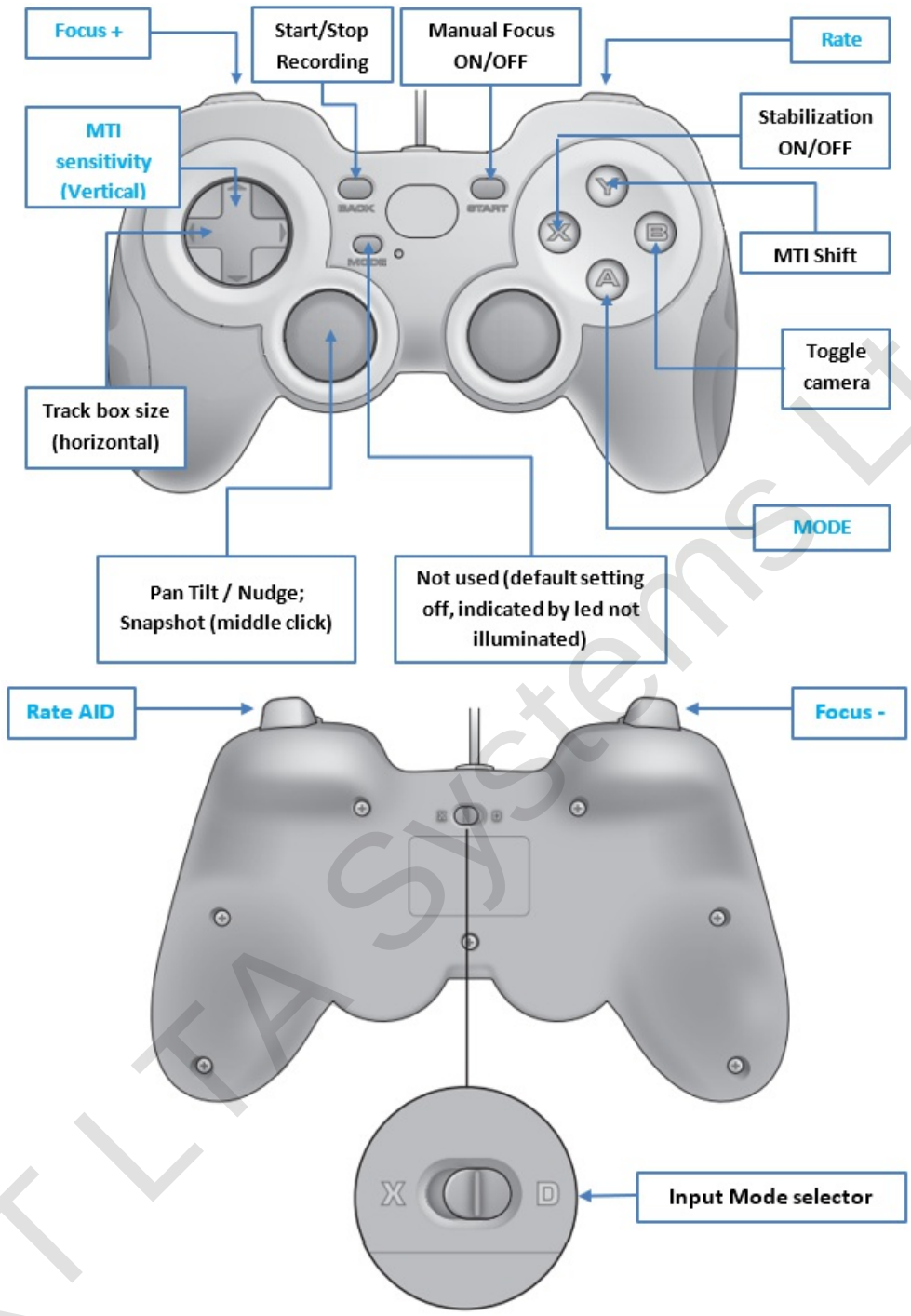
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There are two modes for Joystick buttons. Modes are selected using the **MODE** button (button A on joystick). Mode 2 is enabled by holding the “MODE” button and switched back to mode 1 when the button is released. See pictures below for button description for each mode.



Joystick functions description MODE 1

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Joystick function description MODE 2

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6.4.1 Digital Zoom

Digital zoom up to 4x can be adjusted by pressing **Digital zoom –** or **Digital zoom +** buttons. The default (minimal) value is 1.0 and can be set up to 4.0 by step 0.5.

6.4.2 Focus Control

If focus mode is switched to manual, it can be adjusted by **Focus –** or **Focus +** buttons while holding **Focus adjustment mode** button. Focus adjustment function uses same buttons as zoom control. **Focus adjustment mode** button switches between zoom and focus commands.

6.4.3 Joystick control modes

The joystick that is supplied with the Epsilon is able to work in two interface modes – X Input and Direct Input. You can switch between these two modes by sliding a switch on the bottom of the joystick.

For Epsilon software controls to work properly the Direct Input mode has to be selected (switch flipped to D).



Joystick interface mode in Direct Input position

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IP Desktop joystick

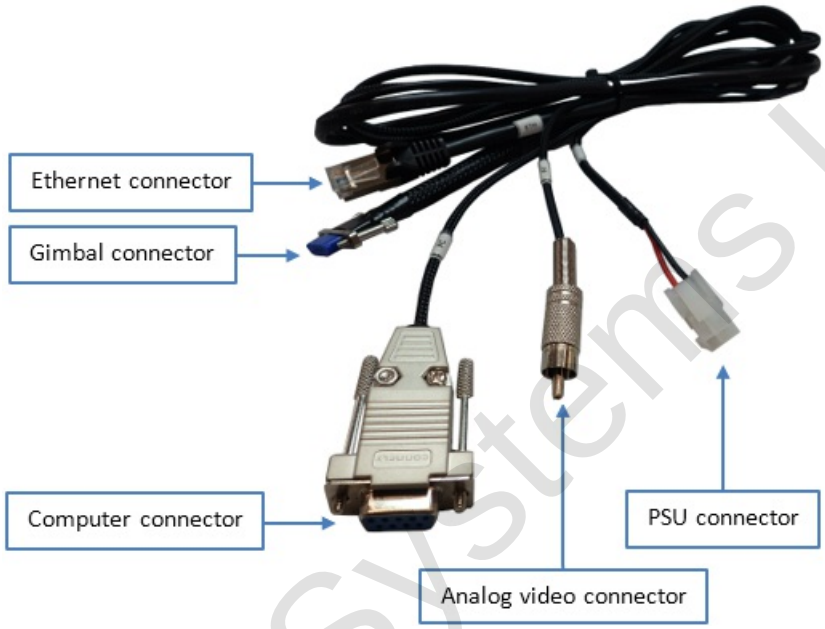
Button	Function
1	Toggle Camera
2	Stabilization on/off
3	Rate mode
4	Rate AID mode
5	Recording on/off
6	Manual focus on/off
7	Digital zoom + / MTI sensitivity +
8	Digital zoom - / MTI sensitivity +
9	MODE (hold the button)
10	Geo mode

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6.5 Quick Setup Kit

6.5.1 Interface harness

The Quick setup kit can be ordered from UAVFACTORY. The kit includes all necessary parts for quick Epsilon integration and start-up. Interface harness has an Epsilon mating connector (either AXON micro D female or D-sub 9 high density male) which is routed to the power supply, RJ45 connector for Ethernet connection, DSUB connector for RS232 communication, and RCA connector for analog video output. RCA analog video output is only available on Epsilon 140, 140Z and 140LC.



Interface harness adapter for Epsilon 140 and 140Z

The Quick setup kit includes the 24V AC/DC power adapter with mating connector for quick Epsilon setup.



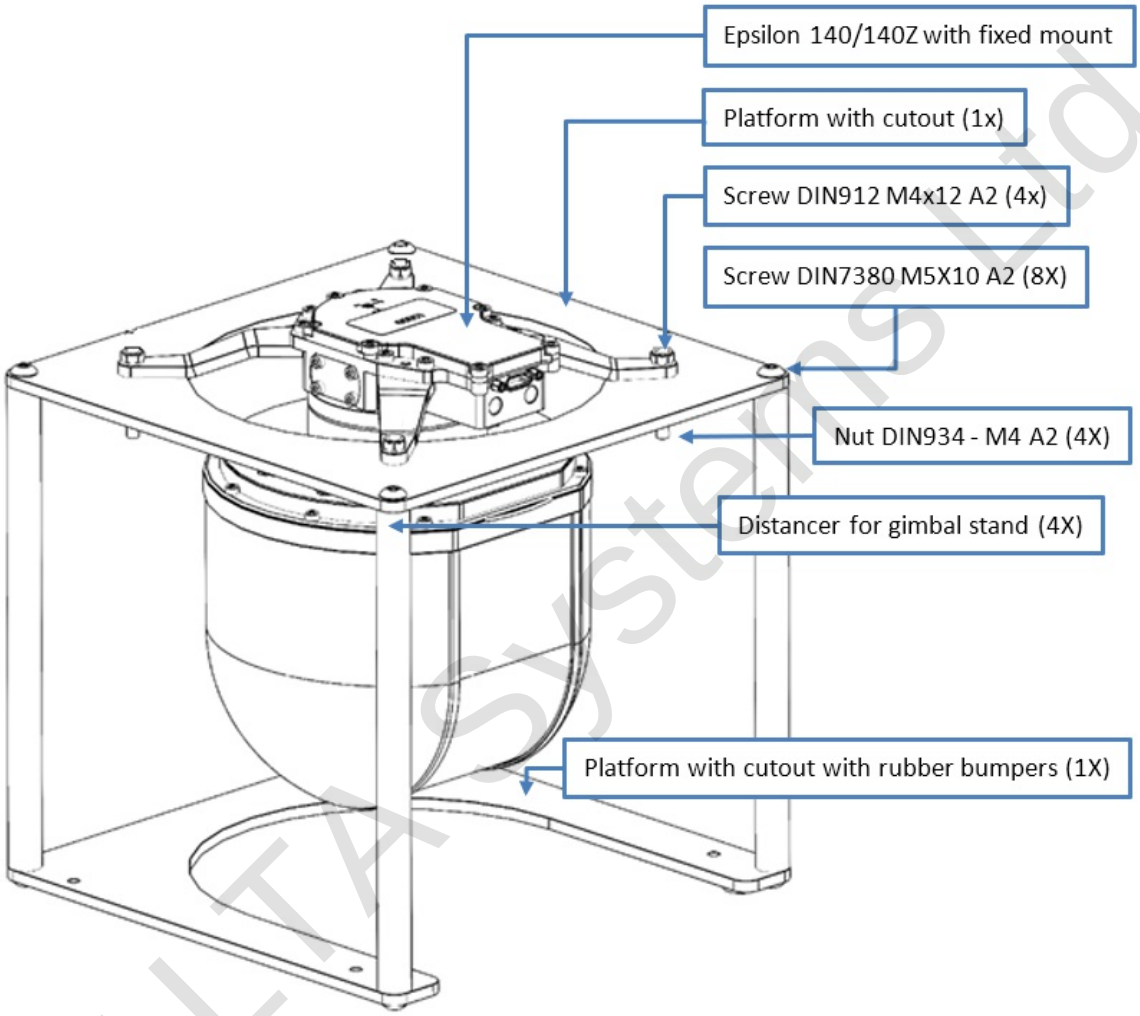
AC/DC power supply

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6.5.2 Aluminum Stand

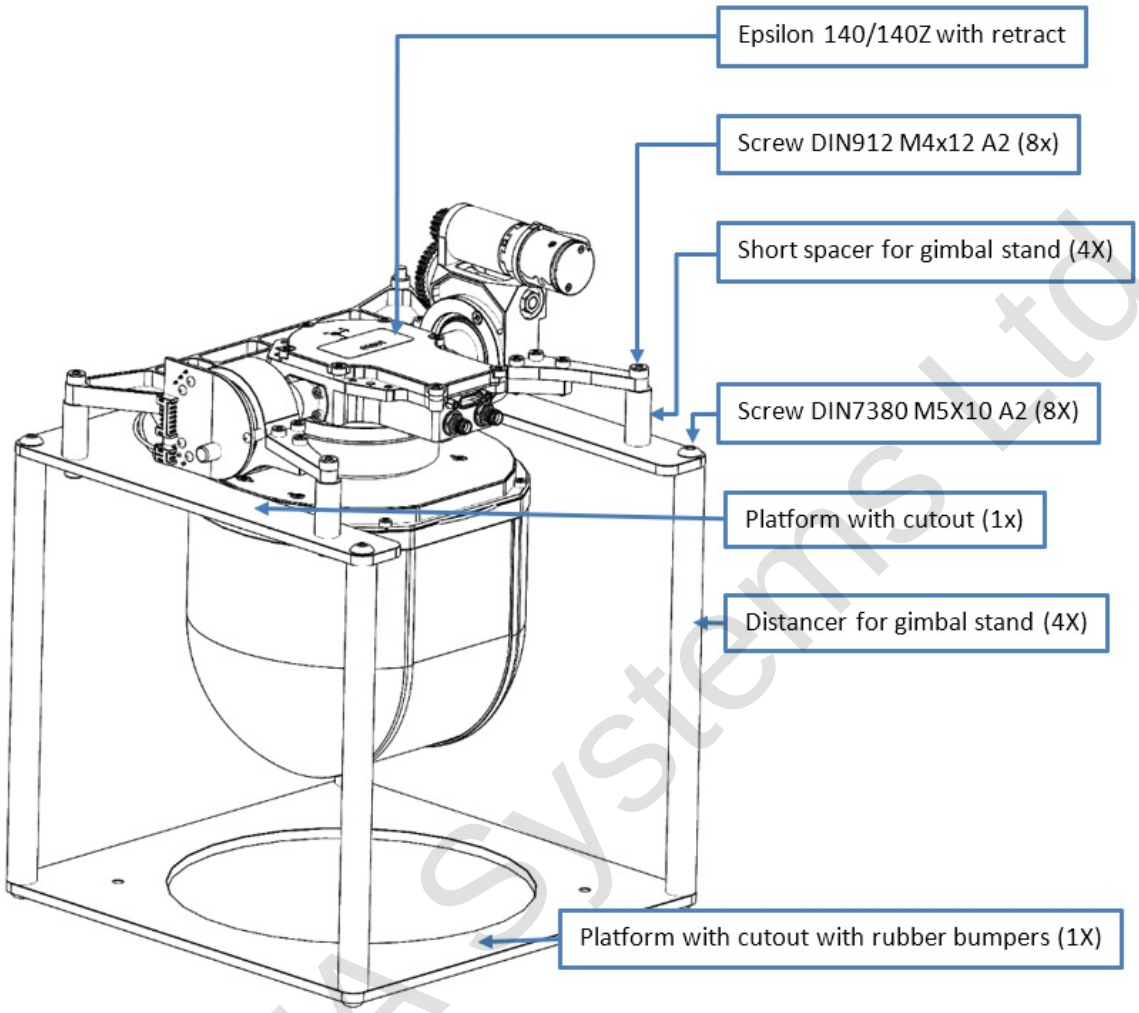
The aluminum stand is used for Epsilon desktop operation before the integration and to perform tests or firmware updates.

6.5.2.1 Epsilon Stand For 140/140Z With Fixed Mount



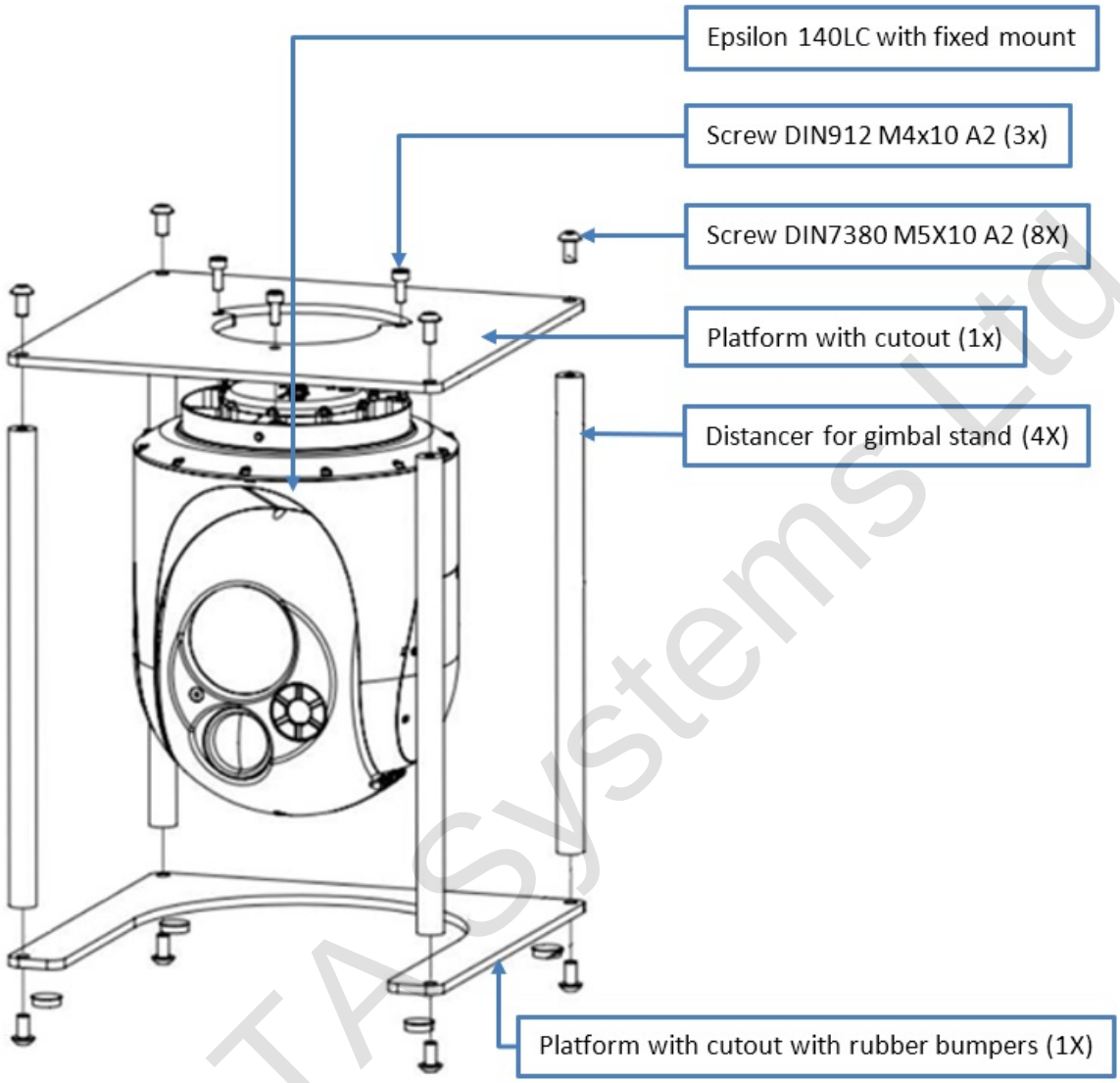
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6.5.2.2 Epsilon Stand For 140/140Z With Retract Mechanism



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6.5.2.3 Stand For Epsilon 140LC/140Z G2



Note: Epsilon 140LC and 140Z G2 has the same mounting pattern.

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7.1 Firmware Update

Firmware update allows to:

- Restore Epsilon system
- Upgrade or add Epsilon features.

NOTE: Make sure you use correct USB-COM adapter to connect Gimbal to PC. As well do not use USB hubs on your PC, Gimbal should be connected using USB to RS232 adapter (Part number ES-U-1001-R10 US232R-10) straight to stand alone USB port on your PC

7.2 Firmware update for Epsilon software version 2.9 and greater

Firmware update steps:

1. Uninstall previous Epsilon software.
2. Install / Extract the new Epsilon software provided by “UAV Factory”.
3. Unzip the new firmware files on your computer provided by “UAV Factory”.
4. Connect Epsilon directly to your computer using serial RS232.
5. Open the Epsilon software.
6. In menu bar select “Window” and choose “Firmware update” option from drop down menu.
7. In “Firmware update” window browse correct “hex” file.
8. Power on the Gimbal (Initializing process will not start).
9. Firmware update process will start and “Status” will show: “Tilt erasing” and after few seconds will show “Tilt writing”.
10. Once the software “Status” shows "Pan writing done. Firmware update is completed!" – Power off the Gimbal.
11. Epsilon is ready for operations.

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7.3 Lens Cleaning

Caution: Do not use any other solvents as they can damage seals around the lens, use only denatured spirit or isopropyl alcohol.

Lenses should be periodically cleaned to ensure the best performance or when they are dirty. To avoid scratching the lens the following order of cleaning must be used:

1. Blow off dirt and dust using rubber air blower
2. Prepare soft tissue paper, soaking it in either denatured spirit or isopropyl alcohol with as high as possible alcohol content (>99%)
3. Fold the soaked tissue paper and slowly drag it from center to the side of lens, for this only the folded edge should be used, making sure that dragging motion is slow enough that the alcohol evaporates instantly behind the tissue and there are it doesn't stay on the glass.
4. Each new pass from center to side must be done using clean part of the tissue, making sure that debris is not dragged over the lens



Correct procedure for lens cleaning

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7.4 Humidity Cartridge

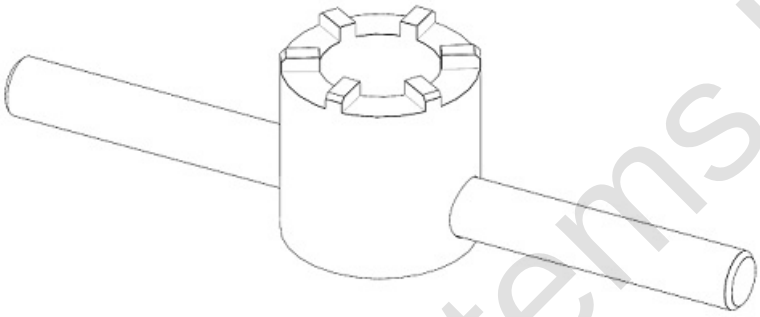
Humidity cartridge prevents condensation, which might damage the electronics.

Humidity cartridge can adsorb about 40 percent of its weight in moisture and can take the relative humidity in a closed container down to about 40 percent. Once saturated, you can replace the Humidity cartridge with new one (ordering from UAV Factory). Humidity cartridge should be replaced if Silica Gel inside Humidity cartridge is dark blue.

All refills are packaged for a shelf life of 12 months when stored in temperate conditions.

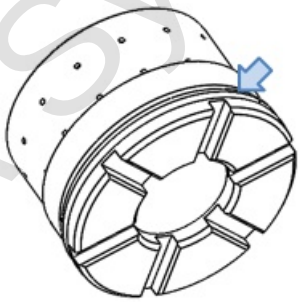
7.4.1 Humidity Cartridge Replacement

Removing and installing of humidity cartridge should only be done with a special tool.



Tool for removing/installing humidity training

The desiccant seal must be lubricated with silicone moisturizing stick!



New Humidity cartridge

Note: During Humidity cartridge installation, tighten Humidity cartridge until it fully rests on payload case structure.
Note: Cotton bud soaked with silicone spray can be used for lubrication.

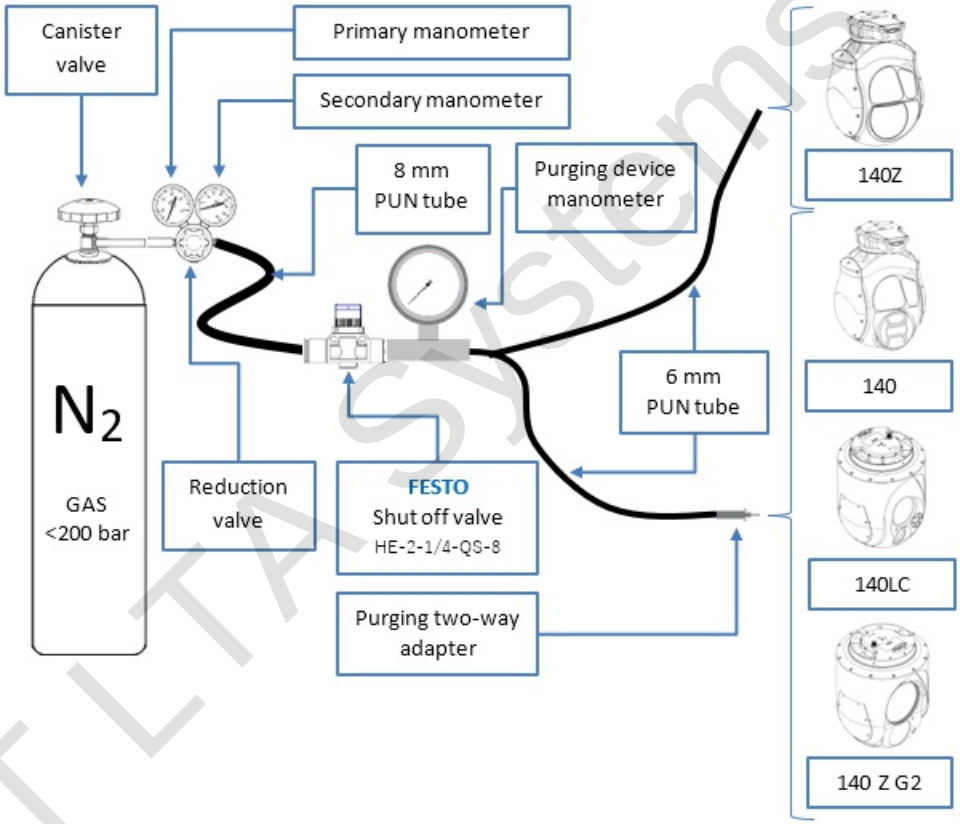
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7.5 Nitrogen Purging

This procedure is meant for purging the humid air out of the gimbal, and fill it with inert gas (99,999% Nitrogen). Without this procedure the humid air inside the camera can cause water condensation on internal components as a result of temperature changes. This can cause blurry image and in the worst case electrical and mechanical component damage and corrosion.
For purging Epsilon 140Z 6 mm PUN tube can be directly inserted in to the port. Epsilon 140,140LC and 140Z G2 have a special purging two-way adapter that should be attached in order to ensure Epsilon ir completely purged. Two-way purging adapter should be attached when purging setup is already adjusted.
Purging procedure should be done very carefully, to avoid any accidental overpressure bursts that could cause damage to the camera.

Caution: Always use eye protection.

7.5.1 General Purging Setup Overview



Purging system setup

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These steps are required to achieve the nominal purging pressure with this setup.

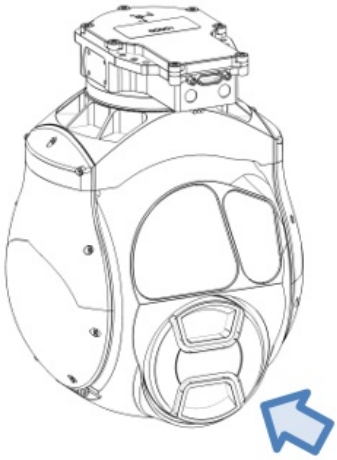
- Make sure that all pneumatic connections are securely fastened.
- Make sure that the reduction valve is closed.
- Make sure that FESTO shut-off valve is open.
- **Leave the output 6mm PUN tube disconnected from gimbal and two-way adapter disconnected from the 6mm PUN tube!**
- Open the canister valve, minimum of one turn, check the direction on the canister valve itself.
- Check the primary manometer of the reduction valve, it shows the canister pressure (<200 bar)
- Start opening the reduction valve. turn it very, very slowly, and adjust the pressure on the **secondary manometer** valve manometer **below 1 bar!**
- Check for gas flow at the output of 6 mm PUN tube, by closing the output tube with a finger and checking the output manometer pressure.
- Once the initial pressure of below 1 bar has been adjusted proceed by fine adjusting the output pressure.
- Adjust the output pressure to **0.25 ± 0.05 bar, use the purging device manometer for reference**. Start closing the reduction valve and repeatedly open and close the output using finger, to let the previous pressure out.
- Attach purging two-way adapter in case of Epsilon 140, Epsilon 140LC and Epsilon 140Z G2.
- **To stop or start the gas flow use only the FESTO shut-off valve, otherwise the nominal pressure adjustment will be lost!**
- **After procedure make sure that Chapter "End of procedure" is completed.**

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7.5.2 EPSILON 140

7.5.2.1 Purging Setup

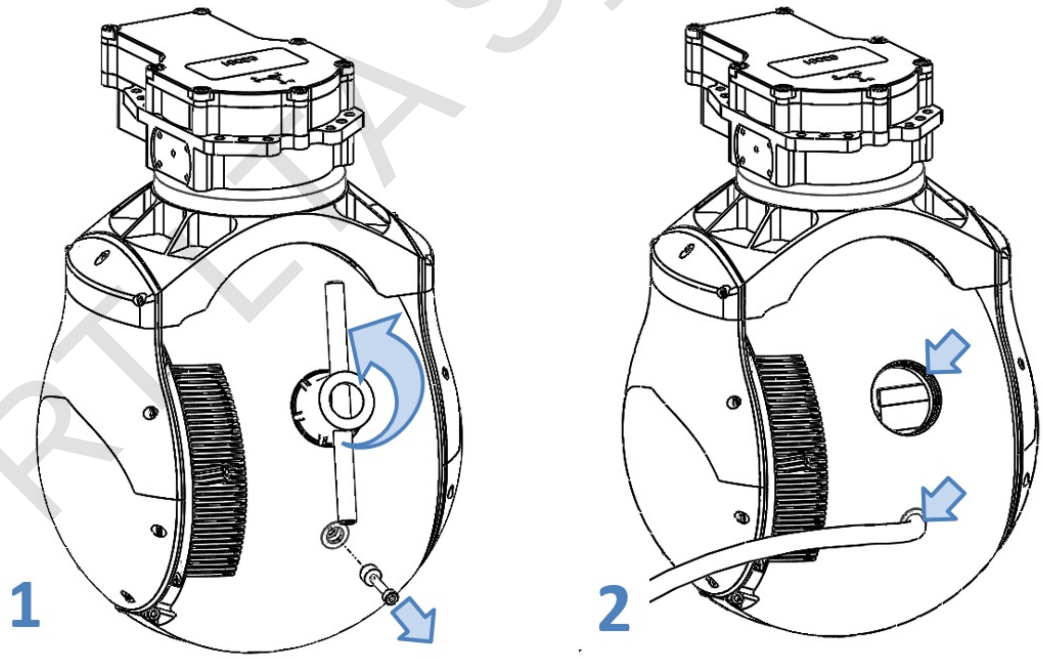
Note: Always use the lens protective cover!



Epsilon 140 lens protective cover

After the purging setup has been adjusted to the required pressure, it is ready to be connected to the Epsilon 140 ball purging port. To connect the purging setup to the camera, do the following:

- Remove the valve bushing cork and screw in a M2,5x16 screw (tighten only with fingers) and pull out the cork.
- Remove desiccant cartridge, with special tool
- Push the purging two-way adapter into the purging port bush, until it hits the bottom of the bush and stops.
- The purging setup is ready for the purging procedure.



Epsilon 140 purging bush cork removal

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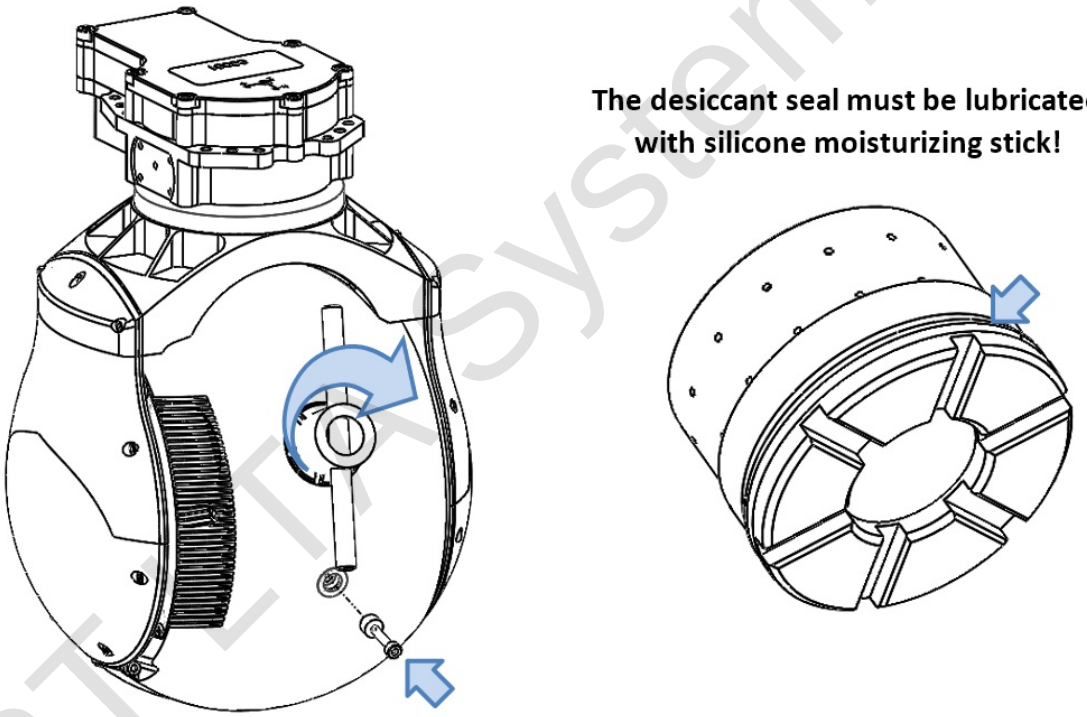
7.5.2.2 Purging Iinitiation And Cycling

- Make sure that all pneumatic connections are securely fastened.

  Open the FESTO Shut off valve to initiate purging. To make sure that the purging process is initiated, check the output manometer. There should be a noticeable drop in pressure, due to constant gas flow.
- Approximately 0,1 bar.

To make sure that no residual air is left in the gimbal, perform this purging procedure 3 times in a row:

- Purge for 1 minute with constant gas flow.
- After that screw in the desiccant cartridge a few turns, without stopping the gas supply, to seal the ball and wait for the gases to stratify.
- After 1 minute, remove the desiccant cartridge again, carefully (the ball is pressurized), and initiate purging.
- Repeat the previous steps 3 times.
- At the end of the third cycle, close the ball with a new and fresh desiccant cartridge (without stopping the gas supply) and by tightening it with the special tool.
- Disconnect the purging two-way adapter and insert the valve cork (remove the screw) into the purging port bush.
- Close the FESTO shut off valve to stop the nitrogen flow.



Epsilon 140 ball closing

The purging procedure is done and the gimbal is inert with nitrogen (<0,3 bar pressure). It is ready for use.

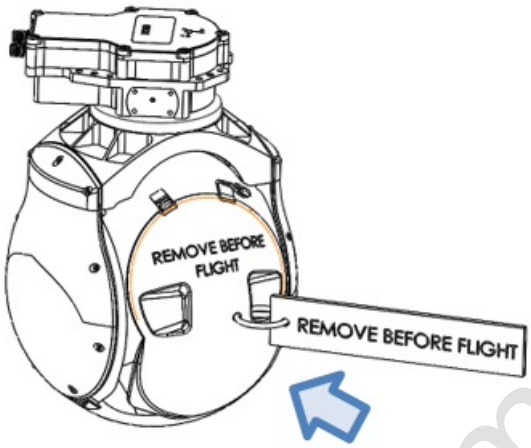
Remember to depressurize the purging system after work. To do that, see paragraph 7.3.3.

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7.5.3 EPSILON 140Z

7.5.3.1 Purging Setup

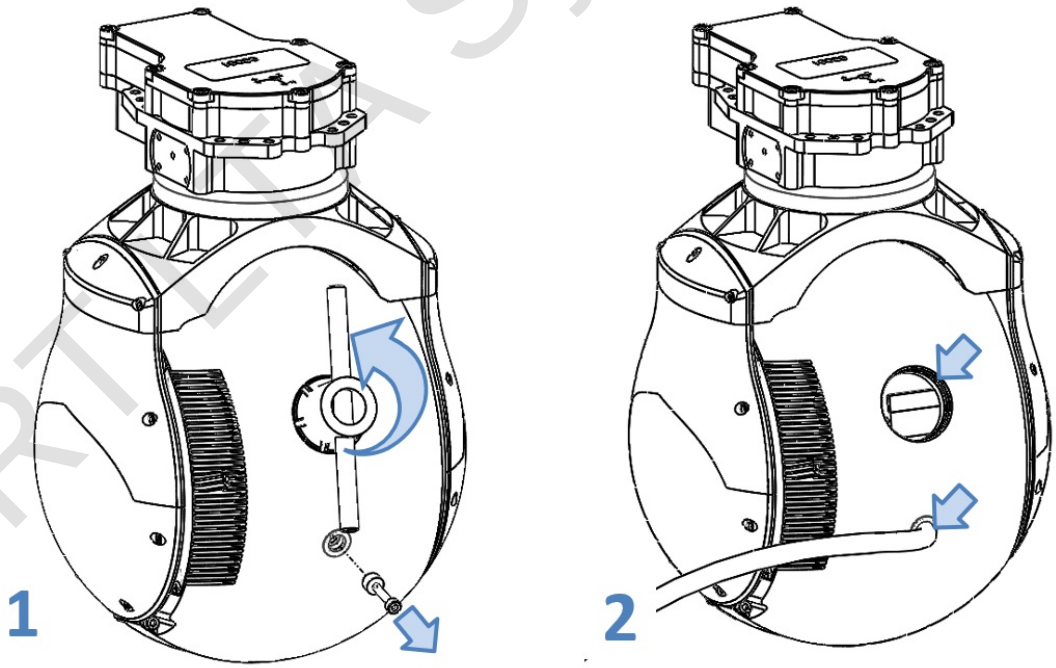
Note: Always use the lens protective cover!



EPSILON 140Z lens protective cover

After the purging setup has been adjusted to the required pressure, it is ready to be connected to the Epsilon 140 ball purging port. To connect the purging setup to the camera do the following:

- Remove the valve bushing cork and screw in an M2,5x16 screw (tighten only with fingers) and pull out the cork.
- Remove the desiccant cartridge, with the special tool
- Push the 6 mm PUN tube into the purging port bush, until it hits the bottom of the bush and stops.
- The purging setup is ready for the purging procedure.



EPSILON 140Z purging bush cork removal

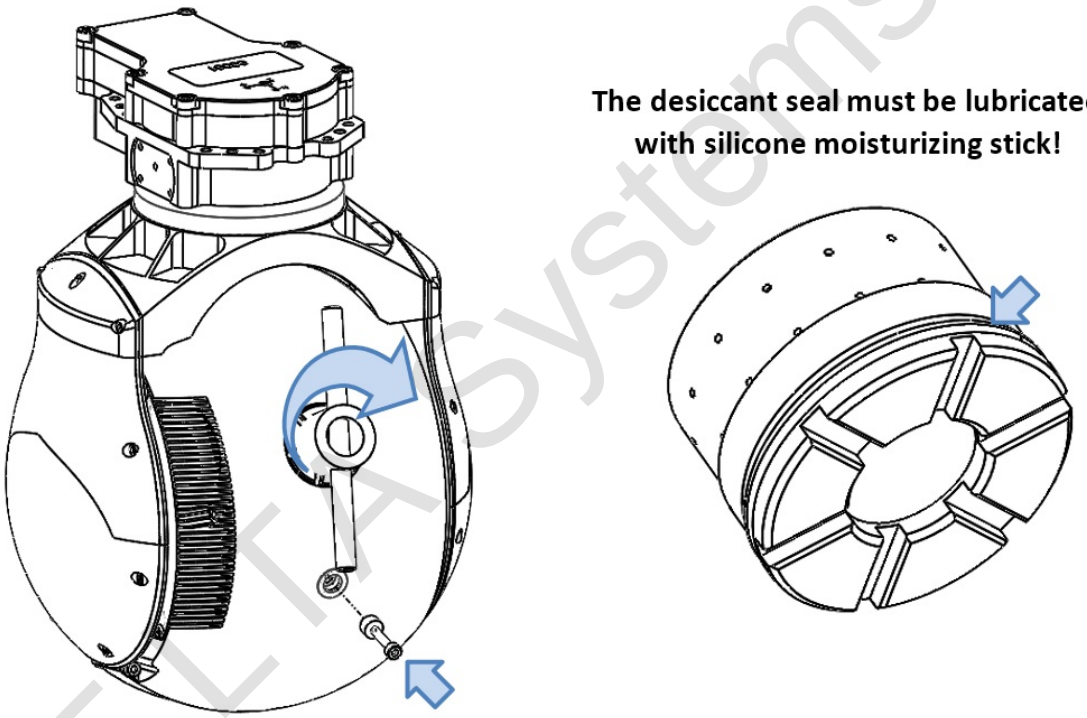
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7.5.3.2 Purging Initiation And Cycling

- Make sure that all pneumatic connections are securely fastened.
- Open the FESTO Shut off valve to initiate purging. To make sure that the purging process is initiated, check the output manometer. There should be a noticeable drop in pressure, due to constant gas flow. Approximately 0,1 bar.

To make sure that no residual air is left in the gimbal, perform this purging procedure 3 times in a row:

- Purge for 1 minute with constant gas flow.
- Close the FESTO shut off valve to stop the nitrogen flow.
- Immediately screw in the desiccant cartridge a few turns to seal the ball and wait 1 minute for the gases to stratify.
- After 1 minute, remove the desiccant cartridge again, and initiate purging by opening the shut off valve.
- Repeat the previous steps 3 times.
- At the end of the third cycle, close the ball with a new and fresh desiccant cartridge, by tightening it with the special tool, disconnect the PUN tube and insert the valve cork (remove the screw) into the purging port bush.



Epsilon 140 ZOOM ball closing

The purging procedure is done and gimbal is inert. It is ready for use.

Remember to depressurize the purging system after work. To do that, see paragraph 7.3.3.

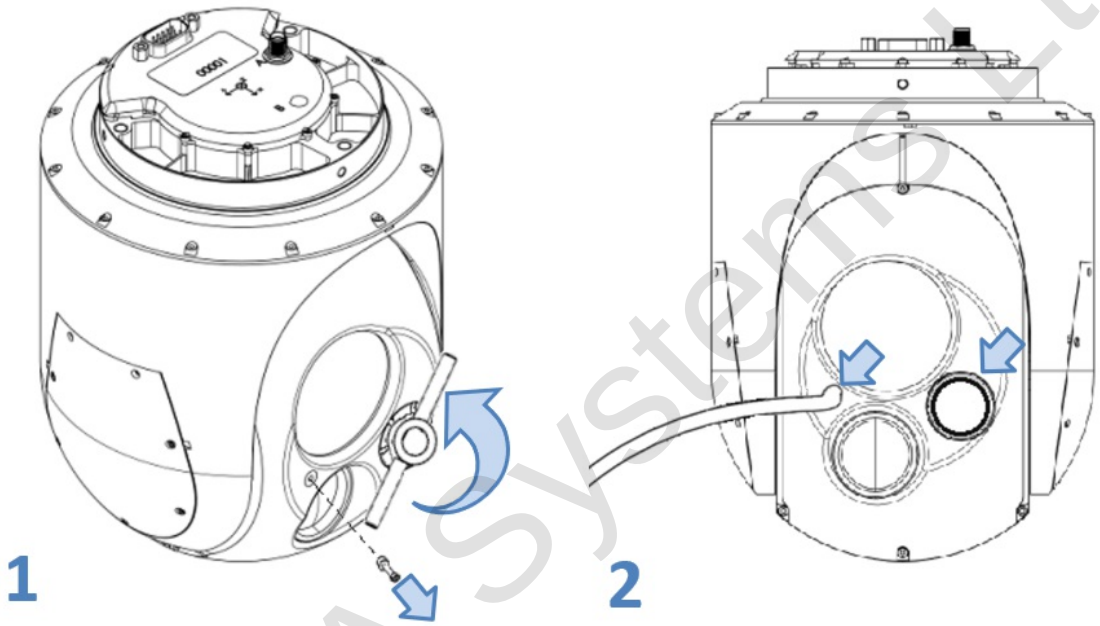
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7.5.4 EPSILON 140LC

7.5.4.1 Purging Setup

After the purging setup has been adjusted to the required pressure, it is ready to be connected to the Epsilon 140 ball purging port. To connect the purging setup to the camera do the following:

- Remove the valve bushing cork and screw in a M2,5x16 screw (tighten only with fingers) and pull out the cork.
- Remove desiccant cartridge, with special tool
- Push the purging two-way adapter into the purging port bush, until it hits the bottom of the bush and stops.
- The purging setup is ready for the purging procedure.



EPSILON 140LC purging bush cork and humidity cartridge removal

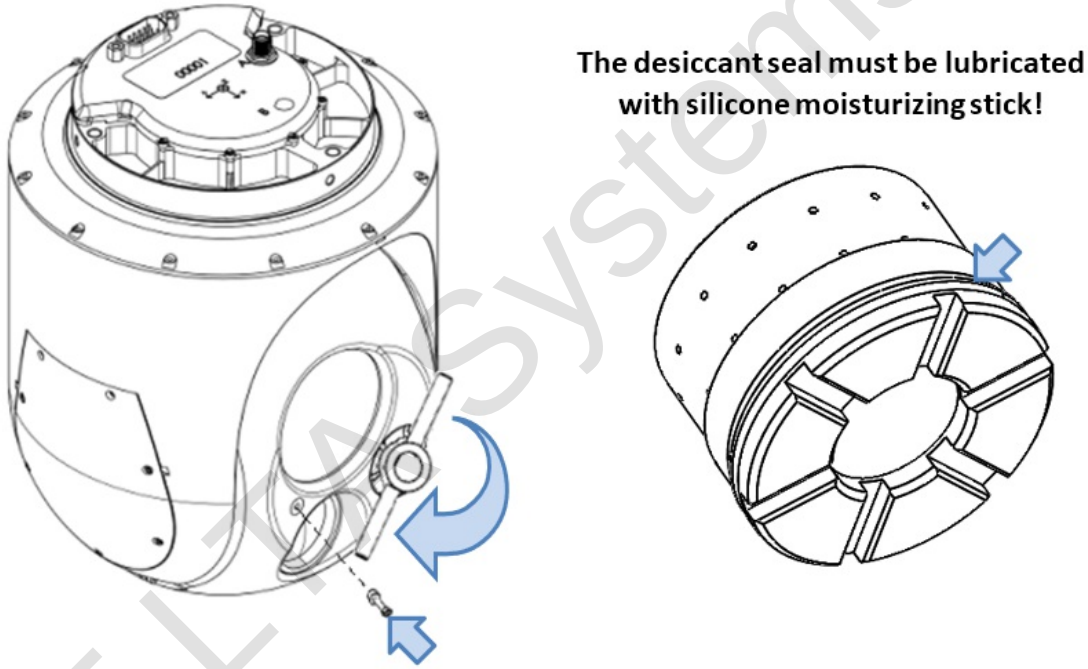
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7.5.4.2 Purging Iinitiation And Cycling

- Make sure that all pneumatic connections are securely fastened.
- Open the FESTO Shut off valve to initiate purging. To make sure that the purging process is initiated, check the output manometer. There should be a noticeable drop in pressure, due to constant gas flow. Approximately 0,1 bar.

To make sure that no residual air is left in the gimbal, perform this purging procedure 3 times in a row:

- Purge for 1 minute with constant gas flow.
- After that screw in the desiccant cartridge a few turns, without stopping the gas supply, to seal the ball and wait for the gases to stratify.
- After 1 minute, remove the desiccant cartridge again, carefully (the ball is pressurized), and initiate purging.
- Repeat the previous steps 3 times.
- At the end of the third cycle, close the ball with a new and fresh desiccant cartridge (without stopping the gas supply) and by tightening it with the special tool.
- Disconnect the purging two-way adapter and insert the valve cork (remove the screw) into the purging port bush.
- Close the FESTO shut off valve to stop the nitrogen flow.



Epsilon 140 ZOOM ball closing

The purging procedure is done and the gimbal is inert. It is ready for use.

Remember to depressurize the purging system after work. To do that, see paragraph 7.3.3.

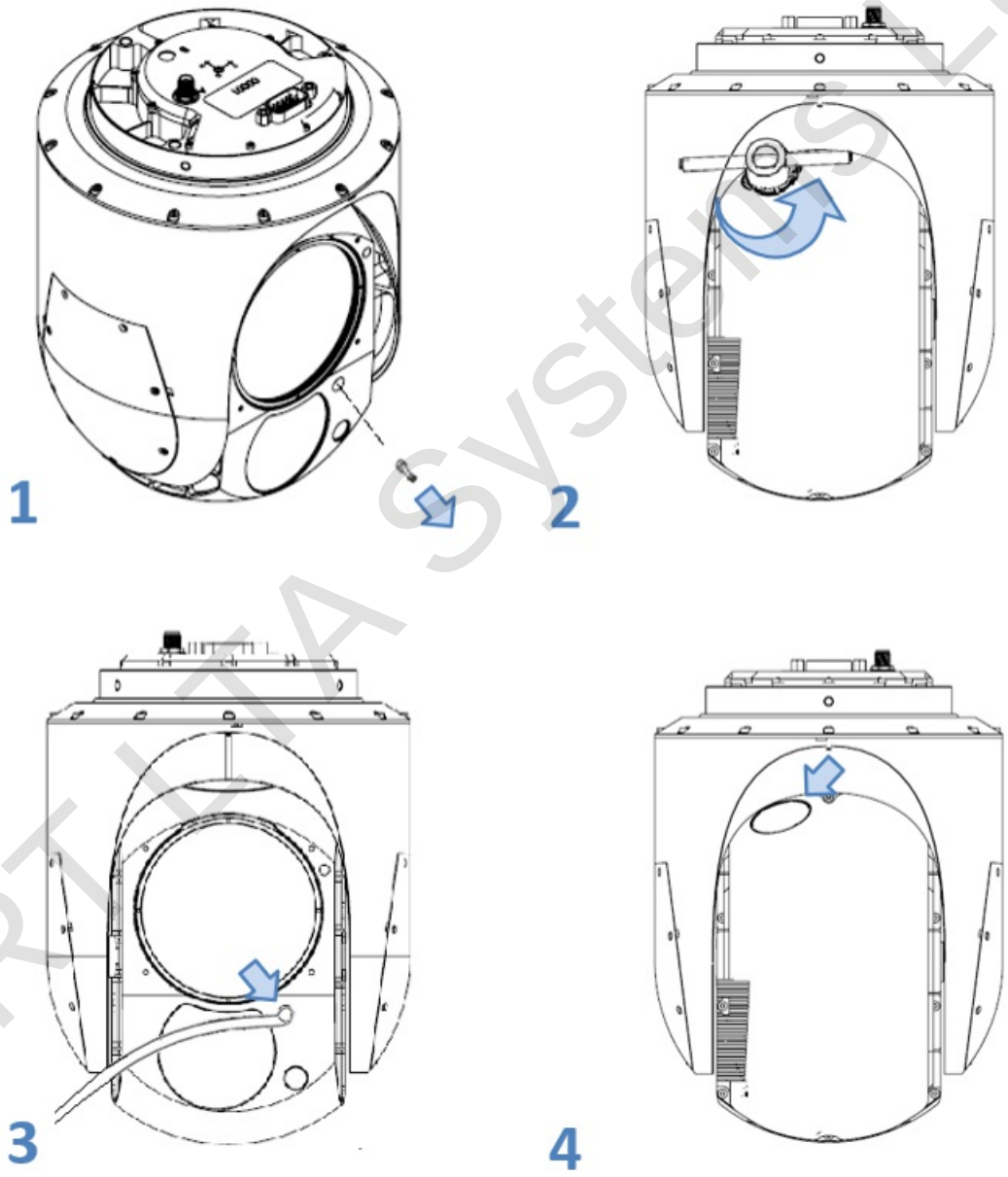
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7.5.5 EPSILON 140Z G2

7.5.5.1 Purging Setup

After the purging setup has been adjusted to the required pressure, it is ready to be connected to the Epsilon 140 ball purging port. To connect the purging setup to the camera do the following:

- Remove the valve bushing cork and screw in a M2,5x16 screw (tighten only with fingers) and pull out the cork.
- Remove desiccant cartridge, with special tool
- Push the purging two-way adapter into the purging port bush, until it hits the bottom of the bush and stops.
- The purging setup is ready for the purging procedure.



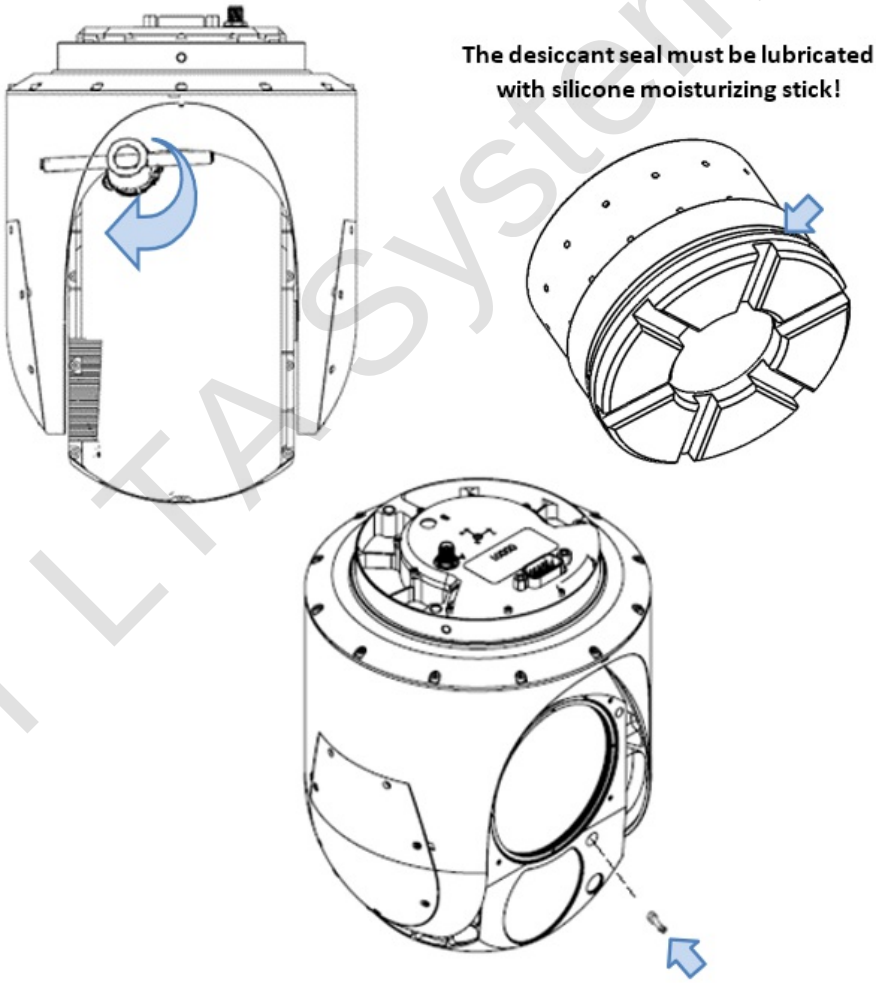
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7.5.5.2 Purging Initiation And Cycling

- Make sure that all pneumatic connections are securely fastened.
- Open the FESTO Shut off valve to initiate purging. To make sure that the purging process is initiated, check the output manometer. There should be a noticeable drop in pressure, due to constant gas flow. Approximately 0,1 bar.

To make sure that no residual air is left in the gimbal, perform this purging procedure 3 times in a row:

- Purge for 1 minute with constant gas flow.
- After that screw in the desiccant cartridge a few turns, without stopping the gas supply, to seal the ball and wait for the gases to stratify.
- After 1 minute, remove the desiccant cartridge again, carefully (the ball is pressurized), and initiate purging.
- Repeat the previous steps 3 times.
- At the end of the third cycle, close the ball with a new and fresh desiccant cartridge (without stopping the gas supply) and by tightening it with the special tool.
- Disconnect the purging two-way adapter and insert the valve cork (remove the screw) into the purging port bush.
- Close the FESTO shut off valve to stop the nitrogen flow.



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7.5.6 End Procedure

To depressurize the purging system, do the following:

- Open the FESTO shut off valve
- Close the reduction valve, by turning CCW.
- Close the canister valve, CW
- Disconnect the manometer from the 8 mm PUN tube
- Release the primary stage pressure, by opening the reduction valve.
- Close the reduction valve.

Note: Both manometers should show zero pressure!

7.5.7 Warnings

- Always use eye protection when working with pressurized systems.
- Be careful not to damage camera glass apertures that are not protected.
- Do not close the canister valve before closing the reduction valve. Otherwise the primary stage of reduction valve will be left pressurized with the canister pressure, that could cause accidental overpressure bursts on later usage, due to active pressure regulation.

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7.6 GPS Antenna Connectors Insulation Application

ATTENTION: Work with Liquid Electrical Tape in a well-vented room!

7.6.1 Applicability

All Epsilon 140/140Z/140LC/140Z G2 Gimbals if GPS Antenna(s) connected.

7.6.2 Tools and Materials

Tools and Materials	
Tool	**Description**
Syringe with Plastic Tip	For Liquid Electrical Tape application
Brush, flat10mm	Alternate Liquid Electrical Tape application tool
Spanner 8mm	GPS Antenna Connector tightening
Material	**Description**
Liquid Electrical Tape, PERFORMIX (P11487)	GPS Antenna Connector insulation
Ethanol	Cleaning and degreasing
Soft wipes	Cleaning and degreasing
Q-tips	Cleaning Connector from inside

7.6.3 Procedure

1. Check GPS Antenna Connector(s) condition (Clean, tightened). If necessary, disconnect using 8mm Spanner, clean with ethanol using Q-tips, and tighten Connector(s);
2. Clean GPS Antenna Connector(s) from outside using ethanol and soft wipes;
3. Apply Liquid Electrical Tape on GPS Antenna Connector(s) to completely seal it as shown in Figure below;
4. Apply Liquid Electrical Tape in several layers- at least two layers must be applied. Cure time between layers is 10 minutes;
5. Check GPS Antenna Connector(s) completely sealed- see Figure below. If necessary, apply more layers of Liquid Electrical Tape;
6. Cure time before use 4h.



GPS Antenna Connector sealed with Liquid Electrical Tape

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